Learning Global Probabilistic Explanations

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Abstract

Interpreting the predictions of complex black-box classifiers remains a central challenge in explainable artificial intelligence. While local explanations clarify individual predictions, there is a significant need for global probabilistic explanations that capture a model's overall behavior across the input distribution. Such an explanation is a small subset K of features, whose relevance is measured by the expected probability, over a random input $\boldsymbol{x}$, that the classifier assigns the same label to $\boldsymbol{x}$ and to a random input agreeing with $\boldsymbol{x}$ on K. In this study, we investigate the task of identifying maximally relevant global explanations under a cardinality constraint, focusing on product distributions with full support on discrete feature spaces. We introduce a spectral explainer that leverages membership queries to the black-box classifier and employs Fourier-analytic, junta-learning methods to produce probably approximately correct (PAC) explanations. Our algorithm is fixed-parameter tractable with respect to the explanation size limit, maximum feature cardinality, and desired accuracy. Experiments on synthetic and real-world datasets confirm the theoretical guarantees and show that the spectral explainer outperforms a greedy baseline as the explanation size increases.

1 Introduction

Classification has long been the cornerstone of machine learning research [37], evolving from early linear models and decision trees to modern deep neural networks, ensemble methods, and graph-based architectures [36]. While maximizing predictive accuracy was once the primary objective, the integration of machine learning into high-stakes, real-world applications [27] has made interpretability and transparency essential. In these contexts, strong performance alone is insufficient—stakeholders demand explanations that are clear, understandable, and well-justified. This imperative has catalyzed rapid advances in *explainable artificial intelligence (XAI)*, a field dedicated to making complex models transparent and accessible to human users [10, 29].

A central question in XAI is to explain *why* a classifier f assigns a specific label $f(\boldsymbol{x})$ to an input instance $\boldsymbol{x}$. Among the many explanation types developed to address this need [13, 34], decision rules or *anchors* [35] stand out for their conceptual clarity. When $\boldsymbol{x}$ is described by a set of d interpretable features, an anchor can be represented by a subset $K \subseteq [d]$ such that altering features outside K is not likely to affect the outcome. Notably, if the restriction $\boldsymbol{x}_K$ of $\boldsymbol{x}$ to K suffices to determine $f(\boldsymbol{x})$ with certainty, K is known as an *abductive explanation* [11, 20] or *sufficient reason* [4, 12].

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Despite their strong theoretical foundations, abductive explanations are frequently too large to be practical [19, 20]. This limitation stems from human cognition: psychological research consistently finds that people can process only a small number of interacting variables—typically four or five—before cognitive overload occurs [17, 22]. Moreover, the complexity of an explanation depends not only on the number of features k, but also on their domain cardinality c: since the number of configurations grows as c^k, the demand on working memory rises exponentially. Consequently, empirical XAI research emphasizes the need for concise explanations to achieve genuine interpretability [26, 31].

Motivated by these cognitive limitations, recent advances in XAI have shifted toward *probabilistic (abductive) explanations*, which aim to strike a balance between accuracy and interpretability [7, 21, 40]. Within this framework, the explanatory power of a feature subset K is measured by its *local relevance*: the probability that a classifier f assigns the same label to a random instance $\boldsymbol{z}$ as to a reference $\boldsymbol{x}$, provided $\boldsymbol{x}$ and $\boldsymbol{z}$ agree on all features in K. Given a user-specified size limit k and a distribution $\mathcal{D}$ over instances, the objective is to find a subset K of size at most k that maximizes:

$$\mathbb{P}_{\boldsymbol{z}\sim\mathcal{D}(\cdot|\boldsymbol{z}_K=\boldsymbol{x}_K)}\left[f(\boldsymbol{x})=f(\boldsymbol{z})\right]. \tag{1}$$

For example, consider a loan approval model f that evaluates applicants based on `Age`, `Income`, and `Debt-to-Income`. Suppose an applicant $\boldsymbol{x}$ is denied a loan: although their debt-to-income ratio is low, their income falls below the model's threshold. In this setting, `Income` acts as a relevant local explanation: for any applicant $\boldsymbol{z}$ with the same income, the model will likely reach the same decision, regardless of age. Thus, knowing income alone suffices to predict $\boldsymbol{x}$'s outcome with high probability.

While prior work has focused on the *local* perspective—explaining individual predictions—recent research increasingly pursues a more ambitious *global* perspective, aiming to capture a model's behavior across the entire input space [5, 6]. In this setting, the feature subset K serves as a *junta* [8, 9], partitioning the input space into regions defined by all possible assignments to K. With a size constraint k and a distribution $\mathcal{D}$, the goal is to find a subset K of size at most k that maximizes:

$$\mathbb{P}_{\boldsymbol{x}\sim\mathcal{D},\ \boldsymbol{z}\sim\mathcal{D}(\cdot|\boldsymbol{z}_K=\boldsymbol{x}_K)}\left[f(\boldsymbol{x})=f(\boldsymbol{z})\right]. \tag{2}$$

This global formulation assesses how consistently the model behaves within each region defined by K. Returning to the loan example, $K=\{\texttt{Income}\}$ might be adequate for one applicant, but may show low global relevance if the model's outcomes depend on feature interactions. In contrast, a pair such as $K=\{\texttt{Income},\texttt{Debt-to-Income}\}$ can partition applicants into groups—such as "High Income, Low Debt"—where decisions are consistently positive. Maximizing (2) thus uncovers the feature combinations that drive reliable model behavior across the input distribution.

In essence, (2) is simply the average of (1) when $\boldsymbol{x}$ is no longer supplied by the user but drawn from $\mathcal{D}$. Yet, this shift fundamentally reshapes the computational landscape for generating explanations.

Recent work by Bassan et al. [6] investigates this distinction by considering the problem of minimizing the size of an explanation subject to a threshold constraint on the relevance error. They show that, under an empirical product distribution $\mathcal{D}$ with constant support size, the global explanation problem is approximable within a constant factor via a greedy algorithm, whereas the local version remains inapproximable. However, in high-dimensional settings, such empirical distributions cover only a negligible fraction of the input space. Consequently, a global explanation K tailored to this sparse support may fail to capture the true logic of f, lacking the look-ahead necessary to detect complex feature interactions that emerge only when a substantial portion of the input space is observed.

By contrast, our results address product distributions that have full support over discrete input spaces, where each feature has cardinality at most c. In this high-support regime, the problem of maximizing (1) subject to a size constraint k is highly intractable for neural networks [40], and remains hard for decision trees [1]. Further evidence shows that, under standard complexity assumptions, minimizing the size of a highly relevant local explanation is not approximable in polynomial time [24, 38, 40], nor is it fixed-parameter tractable in terms of c and k [33].

Despite this local intractability, we demonstrate that near-optimal global explanations can be computed in fixed-parameter tractable time—even when the classifier f is accessed only as a black box. Our framework provides a structural characterization of optimal explanations for (2) via the Fourier spectrum of f for product distributions over discrete domains [32]. Based on this framework, we introduce a *spectral explainer* that combines an extended version of the Kushilevitz–Mansour (KM) procedure [25] for identifying significant Fourier coefficients with the Gopalan–Kalai–Klivans (GKK) pruning strategy [16] for efficient subset selection. Given any accuracy parameter ϵ and confidence

parameter δ, our algorithm returns, with probability at least $1 - \delta$, a feature subset K of size at most k whose global relevance is within ϵ of optimal. The algorithm is fixed-parameter tractable with respect to c, k and $1/\epsilon$, with a runtime polynomial in d, $(ck/\epsilon)^k$, and $\ln(1/\delta)$.

To complement our theoretical findings, we evaluate the spectral explainer on both synthetic and real-world datasets. When the optimal explanation is computable, its relevance closely matches the predicted accuracy, and its query complexity scales with d, k, and ϵ as our analysis suggests. We further benchmark it against a greedy baseline inspired by [6], which lacks guarantees for full-support product distributions. On neural networks, the spectral explainer more frequently finds the optimal explanation, especially as k increases, reflecting the Fourier spectrum's ability to reveal feature interactions missed by greedy selection.

2 Related Work

We say that a subset K is a *k-size τ-relevant* local explanation with respect to a reference instance $\boldsymbol{x}$, a classifier f, and a distribution $\mathcal{D}$ if K contains at most k features and its local relevance (1) is at least τ. Analogous terminology applies to global probabilistic explanations by replacing (1) with (2).

In the local setting, common choices for $\mathcal{D}$ include neighborhood distributions centered at $\boldsymbol{x}$ [35] and the uniform distribution over the input space [21, 23]. Notably, under the uniform distribution, deciding whether a k-size τ-relevant explanation exists for a given $\boldsymbol{x}$ is computationally challenging: the problem is $\mathsf{NP}^{\mathsf{PP}}$-hard for neural networks [40] and NP-hard for decision trees [1]. Even for the special case $\tau = 1$, corresponding to finding k-size abductive explanations, the task is Σ_2^{p}-hard for random forests [3] and neural networks [4]. The prospects for fixed-parameter tractability and approximability remain bleak. Specifically, finding cardinality-minimal τ-relevant local explanations is $\mathsf{W}[2]$-hard with respect to k, even when $\tau = 1$ and f is a monotone neural network [33]. Furthermore, the problem is NP-hard to approximate within any factor of $d^{1-\alpha}$, for any $\alpha > 0$, when f is a neural network [40] or a decision tree [24]. Positive results are limited to relaxed versions of the problem: for linear functions [38], and for arbitrary black-box classifiers under the uniform distribution, at the cost of larger and less precise explanations [7].

The global perspective, as analyzed by Bassan et al. [5, 6], is markedly different from the local setting. Notably, the set function (2) is monotone for any distribution $\mathcal{D}$, and supermodular when $\mathcal{D}$ is a product distribution. As a result, when $\mathcal{D}$ is an empirical product distribution supported on a fixed dataset, a greedy forward selection algorithm approximates cardinality-minimal τ-relevant explanations within a factor that depends on the curvature of (2) and the dataset size.

The dual problem—maximizing the relevance of k-size explanations—closely relates to the well-studied task of learning *k-juntas*, which are functions depending on at most k out of d variables [8, 9]. In its agnostic version, the objective is to find a k-junta h that maximizes the accuracy $\mathbb{P}_{\boldsymbol{z}\sim\mathcal{D}}[h(\boldsymbol{z}) = f(\boldsymbol{z})]$ for a given Boolean function f. Learning k-juntas from random examples under the uniform distribution is notoriously hard, even in the realizable case: the best known algorithms run in time $d^{\alpha k}$ for some constant $\alpha < 1$ [30, 39], and the problem is closely tied to learning noisy parities [14]; see also [2] for juntas under attribute noise. With membership queries, Fourier-analytic methods such as the KM algorithm [25] make the problem tractable: in particular, k-juntas are properly and agnostically learnable under the uniform distribution [16]. Although conceptually related, k-junta learning results cannot be directly transferred to global explainability, since their objectives—accuracy versus relevance—differ, and global explanations correspond to partitioning functions rather than Boolean ones. Finally, the problem of computing global probabilistic explanations shares similarities with *knowledge distillation*, which seeks to learn a global, interpretable surrogate model h of the black-box classifier f (see, e.g., [10] for a survey). Yet again, the goals diverge: surrogate models aim to *recover* the classifier by maximizing accuracy (or fidelity), while global explanations seek to *determine* the classifier's predictions by maximizing relevance.

3 A Spectral Learning Framework

The classifiers under consideration in this study are functions mapping data from a discrete, finite space $\boldsymbol{\Omega} = \Omega_1 \times \cdots \times \Omega_d$ to labels in $\{-1, +1\}$. Each index $i \in [d] = \{1, \ldots, d\}$ specifies an interpretable feature taking values in the alphabet $\Omega_i = \{0, \ldots, c_i - 1\}$. Let $c = \max_{i\in[d]} c_i$ denote the largest alphabet size. A product distribution $\mathcal{D}$ over $\boldsymbol{\Omega}$ is defined by $\mathcal{D} = \mathcal{D}_1 \times \cdots \times \mathcal{D}_d$,

where each $\mathcal{D}_i$ is a probability distribution on Ω_i. Throughout this work, we assume all marginal distributions have *full support*, meaning that $\mathcal{D}_i(a) > 0$ for all $i \in [d]$ and $a \in \Omega_i$.

3.1 Fourier Bases

The core of our framework is the construction of an orthonormal basis of functions with respect to the underlying distribution [32]. Focusing on a single domain $\Omega = \{0, \dots, c-1\}$ with distribution $\mathcal{D}$, let $\mathcal{L}^2(\Omega, \mathcal{D})$ denote the Hilbert space with the inner product $\langle f, g \rangle_{\mathcal{D}} = \mathbb{E}_{x \sim \mathcal{D}}[f(x)g(x)]$. An orthonormal basis consists of c functions $\{\phi_0, \phi_1, \dots, \phi_{c-1}\}$ satisfying $\langle \phi_j, \phi_k \rangle_{\mathcal{D}} = \mathbb{I}[j = k]$. Since $\langle 1, 1 \rangle_{\mathcal{D}} = 1$, the constant function can always serve as the first basis function.

Lemma 1. For any distribution $\mathcal{D}$ with full support over an alphabet Ω of size c, an orthonormal basis for $\mathcal{L}^2(\Omega, \mathcal{D})$ with $\phi_0 = 1$ can be computed in $O(c^2)$ time using the Stieltjes procedure [15].

The Stieltjes procedure treats the values $0, \dots, c-1$ as distinct real nodes; since any choice of distinct nodes yields a valid basis, the procedure applies equally to ordinal and categorical features.

For each feature $i \in [d]$, let $\{\phi_{i,0}, \dots, \phi_{i,c_i-1}\}$ be an orthonormal basis of $\mathcal{L}^2(\Omega_i, \mathcal{D}_i)$ with $\phi_{i,0} = 1$. By taking the tensor product of these local bases, we derive a global basis for the space $\mathcal{L}^2(\boldsymbol{\Omega}, \mathcal{D})$. Let $\mathcal{S} = \prod_{i=1}^{d} \{0, 1, \dots, c_i - 1\}$ denote the set of multi-indices, referred to as *coordinates*. The *support* of a coordinate $\boldsymbol{s} \in \mathcal{S}$ is given by $\mathrm{supp}(\boldsymbol{s}) = \{i \in [d] : s_i \neq 0\}$, and its *global basis function* is defined as $\phi_{\boldsymbol{s}}(\boldsymbol{x}) = \prod_{i=1}^{d} \phi_{i,s_i}(x_i)$.

Lemma 2. Let $\mathcal{D}$ be a product distribution with full support over $\boldsymbol{\Omega}$. The set $\{\phi_{\boldsymbol{s}} : \boldsymbol{s} \in \mathcal{S}\}$ forms a complete orthonormal basis for $\mathcal{L}^2(\boldsymbol{\Omega}, \mathcal{D})$. As a result, any function $f : \boldsymbol{\Omega} \to \mathbb{R}$ admits a unique Fourier expansion:

$$f(\boldsymbol{x}) = \sum_{\boldsymbol{s} \in \mathcal{S}} \hat{f}(\boldsymbol{s}) \phi_{\boldsymbol{s}}(\boldsymbol{x}) \quad \text{where} \quad \hat{f}(\boldsymbol{s}) = \langle f, \phi_{\boldsymbol{s}} \rangle_{\mathcal{D}} = \mathbb{E}_{\boldsymbol{x} \sim \mathcal{D}}[f(\boldsymbol{x}) \phi_{\boldsymbol{s}}(\boldsymbol{x})].$$

For any $p \in [1, \infty)$, the functional p-norm and the spectral p-norm of a function f are defined as:

$$\|f\|_p = (\mathbb{E}_{\boldsymbol{x} \sim \mathcal{D}}[|f(\boldsymbol{x})|^p])^{1/p} \quad \text{and} \quad L_p(f) = \left(\sum_{\boldsymbol{s} \in \mathcal{S}} |\hat{f}(\boldsymbol{s})|^p \right)^{1/p}.$$

We also use $L_\infty(f) = \max_{\boldsymbol{s} \in \mathcal{S}} |\hat{f}(\boldsymbol{s})|$ and $L_0(f) = |\mathrm{supp}(f)|$, where $\mathrm{supp}(f)$ is the spectral support of f, that is, the set of coordinates $\boldsymbol{s}$ for which $\hat{f}(\boldsymbol{s}) \neq 0$. As a corollary of Lemma 2, both the Plancherel theorem, $\langle f, g \rangle_{\mathcal{D}} = \sum_{\boldsymbol{s} \in \mathcal{S}} \hat{f}(\boldsymbol{s}) \hat{g}(\boldsymbol{s})$, and Parseval's identity, $\|f\|_2 = L_2(f)$, hold in this general setting. In particular, $\|f\|_2^2 = 1$ when f takes values in $\{-1, +1\}$.

The *total energy* $W(f)$ of a function $f : \boldsymbol{\Omega} \to \mathbb{R}$ is given by $W(f) = L_2(f)^2$. More generally, the *energy* $W_{\mathcal{T}}(f)$ of f on a subset $\mathcal{T} \subseteq \mathcal{S}$ is the restriction of $W(f)$ to the coordinates in $\mathcal{T}$:

$$W_{\mathcal{T}}(f) = \sum_{\boldsymbol{s} \in \mathcal{T}} \hat{f}(\boldsymbol{s})^2.$$

3.2 The Integer Kushilevitz–Mansour Algorithm

Building upon the original Kushilevitz–Mansour (KM) algorithm [25], which identifies large Fourier coefficients of Boolean functions on the uniform hypercube, we introduce an extended version tailored for product distributions over general discrete domains.

Given a coordinate $\boldsymbol{s} \in \mathcal{S}$, let $\boldsymbol{s}_{p:q}$ denote the sequence $(s_p, s_{p+1}, \dots, s_q)$ for integers $1 \leq p \leq q \leq d$. The p-prefix of $\boldsymbol{s}$ comprises its first p indices, with the 0-prefix defined as the empty sequence $\boldsymbol{\lambda}$. For a prefix $\boldsymbol{\sigma}$ of length p, we write $\phi_{\boldsymbol{\sigma}}(\boldsymbol{x}_{1:p}) = \prod_{i=1}^{p} \phi_{i,\sigma_i}(x_i)$, with $\phi_{\boldsymbol{\lambda}} = 1$. The *weight* of a function f relative to $\boldsymbol{\sigma}$ is the energy concentrated on the prefix cone $\mathcal{C}_{\boldsymbol{\sigma}} = \{\boldsymbol{s} \in \mathcal{S} : \boldsymbol{s}_{1:p} = \boldsymbol{\sigma}\}$. Hereafter, we write $W_{\boldsymbol{\sigma}}$ instead of $W_{\mathcal{C}_{\boldsymbol{\sigma}}}(f)$ when the context is clear. As in the Boolean case, this weight can be expressed as an expectation over correlated pairs of inputs:

$$W_{\boldsymbol{\sigma}} = \mathbb{E}_{\boldsymbol{u}, \boldsymbol{u}' \sim \mathcal{D}_{1:p}, \boldsymbol{v} \sim \mathcal{D}_{p+1:d}} \big[f(\boldsymbol{u}, \boldsymbol{v}) \, \phi_{\boldsymbol{\sigma}}(\boldsymbol{u}) \, f(\boldsymbol{u}', \boldsymbol{v}) \, \phi_{\boldsymbol{\sigma}}(\boldsymbol{u}') \big], \tag{3}$$

Algorithm 1: Integer Kushilevitz–Mansour (IKM) Algorithm

Input: Membership oracle for f, product distribution $\mathcal{D}$, threshold θ, confidence δ
Output: A sparse hypothesis $g : \mathbf{\Omega} \to \mathbb{R}$

Function `IKM(`f, θ, δ`):`
- $g(\boldsymbol{x}) \leftarrow 0$, $c \leftarrow \max_i c_i$
- $\delta_0 \leftarrow \theta^2\delta/(8dc)$
- $b \leftarrow \lceil 8\ln(1/\delta_0)\rceil$, $n \leftarrow \lceil 64/\theta^4\rceil$
- `Search(`$\boldsymbol{\lambda}$`)`
- **return** $g(\boldsymbol{x})$

Function `Search(`$\boldsymbol{\sigma}$`):`
- $\hat{W}_{\boldsymbol{\sigma}} \leftarrow$ `Weight(`$f, \boldsymbol{\sigma}$`)`
- **if** $\hat{W}_{\boldsymbol{\sigma}} \geq \theta^2/2$ **then**
  - **if** $\mathrm{size}(\boldsymbol{\sigma}) = d$ **then**
    - $\boldsymbol{s} \leftarrow \boldsymbol{\sigma}$
    - $\hat{g}(\boldsymbol{s}) \leftarrow$ `Coefficient(`$f, \boldsymbol{s}$`)`
    - $g(\boldsymbol{x}) \leftarrow g(\boldsymbol{x}) + \hat{g}(\boldsymbol{s})\phi_{\boldsymbol{s}}(\boldsymbol{x})$
  - **else**
    - $p \leftarrow \mathrm{size}(\boldsymbol{\sigma})$
    - **for** $j = 0$ **to** $c_{p+1} - 1$ **do**
      - `Search(`$(\boldsymbol{\sigma}, j)$`)`

Function `Weight(`$f, \boldsymbol{\sigma}$`):`
- $p \leftarrow \mathrm{size}(\boldsymbol{\sigma})$
- **for** $j = 1$ **to** b **do**
  - Sample $\boldsymbol{U}, \boldsymbol{U}' \sim \mathcal{D}^n_{1:p}$ and $\boldsymbol{V} \sim \mathcal{D}^n_{p+1:d}$
  - **for** $i = 1$ **to** n **do**
    - $r_i \leftarrow f(\boldsymbol{U}_i, \boldsymbol{V}_i)\,\phi_{\boldsymbol{\sigma}}(\boldsymbol{U}_i)$
    - $r'_i \leftarrow f(\boldsymbol{U}'_i, \boldsymbol{V}_i)\,\phi_{\boldsymbol{\sigma}}(\boldsymbol{U}'_i)$
  - $\bar{W}_j \leftarrow \frac{1}{n}\sum_{i=1}^n r_i r'_i$
- **return** $\mathrm{median}(\bar{W}_1, \ldots, \bar{W}_b)$

Function `Coefficient(`$f, \boldsymbol{s}$`):`
- **for** $j = 1$ **to** b **do**
  - Sample $\boldsymbol{Z} \sim \mathcal{D}^n$
  - **for** $i = 1$ **to** n **do**
    - $w_i \leftarrow f(\boldsymbol{Z}_i)\,\phi_{\boldsymbol{s}}(\boldsymbol{Z}_i)$
  - $\bar{C}_j \leftarrow \frac{1}{n}\sum_{i=1}^n w_i$
- **return** $\mathrm{median}(\bar{C}_1, \ldots, \bar{C}_b)$

Estimating (3) requires evaluating f on pairs of inputs that share the same suffix $\boldsymbol{v}$, which is where membership queries are needed.

As shown in Algorithm 1, the Integer Kushilevitz–Mansour (IKM) algorithm recursively searches for prefixes $\boldsymbol{\sigma}$ whose estimated weight $\hat{W}_{\boldsymbol{\sigma}}$ is at least $\theta^2/2$. If such a prefix has length d, the corresponding Fourier coefficient is estimated and added to the hypothesis. Otherwise, the search continues by extending the prefix with all possible values for the next index.

To ensure the efficiency of the IKM algorithm across non-uniform distributions, we employ the *Median-of-Means* (MoM) estimator [28], that provides sub-Gaussian concentration guarantees based solely on variance bounds. This matters here because basis functions may take large values: $|\phi_{i,j}|$ can be as large as $1/\sqrt{\min_a \mathcal{D}_i(a)}$, so range-based bounds such as Hoeffding's inequality would deteriorate with the smallest marginal probabilities, whereas variances remain bounded.

Lemma 3. Let $f : \mathbf{\Omega} \to \{-1, +1\}$ be a classifier and $\mathcal{D}$ a product distribution with full support. For any prefix $\boldsymbol{\sigma}$ and coordinate $\boldsymbol{s}$, the random variables averaged by WEIGHT and COEFFICIENT have expectations $W_{\boldsymbol{\sigma}}$ and $\hat{f}(\boldsymbol{s})$, respectively, and variance at most 1. Consequently, with $b \geq 8\ln(1/\delta_0)$ blocks of $n \geq 64/\theta^4$ samples each, every call to WEIGHT or COEFFICIENT returns an estimate within $\theta^2/4$ of its target with probability at least $1 - \delta_0$.

The logical structure of the search ensures that IKM recovers all significant spectral energy while producing a sparse hypothesis. Proposition A.2 in the Appendix establishes the completeness and soundness of this strategy: when all estimates are accurate, the search retains every prefix whose cone contains a coefficient with $\hat{f}(\boldsymbol{s})^2 \geq \theta^2$, and only prefixes with weight at least $\theta^2/4$. This leads to our main convergence result:

Theorem 1. Let $f : \mathbf{\Omega} \to \{-1, +1\}$ be a classifier and $\mathcal{D}$ a product distribution with full support and maximum alphabet size c. Assume that each membership query to f and each draw from a marginal $\mathcal{D}_i$ take $O(1)$ time. Then, for any threshold $\theta \in (0, 1]$ and confidence level $\delta \in (0, 1)$, the IKM algorithm makes $O\big(\frac{dc}{\theta^6}\ln\frac{dc}{\theta\delta}\big)$ membership queries, runs in time $O\big(dc^2 + \frac{d^2c}{\theta^6}\ln\frac{dc}{\theta\delta}\big)$, and returns a hypothesis g such that, with probability at least $1 - \delta$:

$$L_\infty(f - g) \leq \theta \quad \text{and} \quad L_0(g) \leq \frac{4}{\theta^2}.$$

A key aspect of the IKM algorithm's efficiency is its handling of the alphabet size c. Although the coordinate space $\mathcal{S}$ can contain up to c^d coordinates, the runtime remains polynomial in d and c. This tractability arises from spectral pruning: prefix cones at a given depth are disjoint, so by Parseval's identity, at most $4/\theta^2$ prefixes of weight at least $\theta^2/4$ can be retained at each depth, independently of c. Each retained prefix spawns at most c children, so at most $4c/\theta^2$ prefixes are examined at each depth, and at most $1 + 4dc/\theta^2$ in total, which explains why the number of queries is nearly linear in c. Moreover, the prefixes of a given depth can all be estimated on a shared batch of samples: the analysis is unchanged, and the number of queries drops to $(2d+1)\, b\, n = O\big(\frac{d}{\theta^4} \ln \frac{dc}{\theta\delta}\big)$, independently of the number of examined prefixes (see Remark A.1 in Appendix A).

4 A Spectral Explanation Algorithm

In this section, we present the foundations of our explanation method. We begin by establishing a structural identity, revealing that the global relevance of any feature subset is precisely quantified by its Fourier energy. Leveraging this characterization, we introduce a spectral explainer that integrates the IKM algorithm with a pruning strategy, inspired by [16], to identify near-optimal explanations.

4.1 Spectral Characterization of Global Relevance

Given an instance $\boldsymbol{x} \in \boldsymbol{\Omega}$ and a subset $K \subseteq [d]$, we denote by $\boldsymbol{x}_K$ the restriction of $\boldsymbol{x}$ to the features in K. We say that two instances $\boldsymbol{x}$ and $\boldsymbol{z}$ *agree* on K if $\boldsymbol{x}_K = \boldsymbol{z}_K$. For a classifier $f : \boldsymbol{\Omega} \to \{-1, +1\}$ and a distribution $\mathcal{D}$ over $\boldsymbol{\Omega}$, the *global relevance error* of K is defined as:

$$\mathsf{err}_{f,\mathcal{D}}(K) = \mathbb{P}_{\boldsymbol{x}\sim\mathcal{D},\ \boldsymbol{z}\sim\mathcal{D}(\cdot|\boldsymbol{z}_K=\boldsymbol{x}_K)}\left[f(\boldsymbol{x}) \neq f(\boldsymbol{z})\right], \tag{4}$$

that is, one minus the global relevance (2). In line with Bassan et al. [6], this error is the average, over $\boldsymbol{x} \sim \mathcal{D}$, of the local relevance error of K at $\boldsymbol{x}$. The global relevance of K can be characterized entirely in terms of the Fourier spectrum of f. Specifically, let f_K denote the *projection* of f onto the feature subset K, defined as:

$$f_K(\boldsymbol{x}) = \mathbb{E}_{\boldsymbol{z}\sim\mathcal{D}}[f(\boldsymbol{z}) \mid \boldsymbol{z}_K = \boldsymbol{x}_K] = \sum_{\boldsymbol{s}\in\mathcal{S}_K} \hat{f}(\boldsymbol{s})\phi_{\boldsymbol{s}}(\boldsymbol{x}),$$

where $\mathcal{S}_K = \{\boldsymbol{s} \in \mathcal{S} : \mathrm{supp}(\boldsymbol{s}) \subseteq K\}$ is the set of coordinates supported on K. The second equality holds because, under a product distribution, $\mathbb{E}[\phi_{\boldsymbol{s}}(\boldsymbol{z}) \mid \boldsymbol{z}_K = \boldsymbol{x}_K]$ equals $\phi_{\boldsymbol{s}}(\boldsymbol{x})$ if $\mathrm{supp}(\boldsymbol{s}) \subseteq K$, and 0 otherwise, since ϕ_{i,s_i} has zero mean whenever $s_i \neq 0$. By Parseval's identity, the total energy of f_K is the energy of f on $\mathcal{S}_K$, that is, $W(f_K) = W_{\mathcal{S}_K}(f)$.

Lemma 4. For any classifier $f : \boldsymbol{\Omega} \to \{-1, +1\}$, any product distribution $\mathcal{D}$ with full support on $\boldsymbol{\Omega}$, and any feature subset K, the global relevance error of K satisfies:

$$\mathsf{err}_{f,\mathcal{D}}(K) = \frac{1}{2}\left(1 - W(f_K)\right).$$

Proof. Let $\boldsymbol{x} \sim \mathcal{D}$ and $\boldsymbol{z} \sim \mathcal{D}(\cdot \mid \boldsymbol{z}_K = \boldsymbol{x}_K)$. Since f takes values in $\{-1, +1\}$, $\mathbb{I}[f(\boldsymbol{x}) \neq f(\boldsymbol{z})] = (1 - f(\boldsymbol{x})f(\boldsymbol{z}))/2$. Conditioning on $\boldsymbol{x}$ gives $\mathbb{E}[f(\boldsymbol{z}) \mid \boldsymbol{x}] = f_K(\boldsymbol{x})$, so by Plancherel's theorem, $\mathbb{E}[f(\boldsymbol{x})f(\boldsymbol{z})] = \langle f, f_K\rangle_{\mathcal{D}} = \sum_{\boldsymbol{s}\in\mathcal{S}_K} \hat{f}(\boldsymbol{s})^2 = W(f_K)$. □

Equivalently, the global relevance of K is $(1+W(f_K))/2$, so finding a maximally relevant explanation of size at most k amounts to finding a subset K with $|K| \leq k$ that maximizes the energy $W(f_K)$. Note that, when $c_i > 2$ for some i, the local bases are not unique, so individual Fourier coefficients depend on the chosen bases. The global relevance of K, however, does not: $W(f_K)$ is the squared spectral 2-norm of the conditional expectation f_K, which is defined without reference to any basis.

4.2 Spectral Recovery, Feature Pruning, and Subset Selection

By carefully calibrating the precision of the IKM phase, we ensure that the spectral energy of the hypothesis g serves as a reliable proxy for the energy of f across all possible explanations of size k. The following lemma is based on the observation that the projection f_K has at most $|\mathcal{S}_K| = \prod_{i\in K} c_i \leq c^k$ coefficients.

Algorithm 2: Spectral Explainer (SE)

Input: Membership query access to classifier f, product distribution $\mathcal{D}$, size limit k, accuracy ϵ, confidence δ.

Set $\epsilon' \leftarrow 2\epsilon/7$
`// Spectral Recovery`
Run IKM on f with threshold $\theta = \epsilon'/c^{k/2}$ and confidence δ to obtain g
`// Feature Pruning`
Identify the influence set $I = \{i \in [d] : \mathrm{Inf}_i^k[g] \geq \epsilon'/k\}$
`// Subset Selection`
Find $K \subseteq I$ with $|K| \leq k$ that maximizes the projected energy $W(g_K)$

return K

Lemma 5. Let $f : \mathbf{\Omega} \to \{-1, +1\}$ be a classifier and $g : \mathbf{\Omega} \to \mathbb{R}$ such that $L_\infty(f - g) \leq \epsilon/c^{k/2}$ for some $\epsilon \in (0, 1)$. Then, for any subset $K \subseteq [d]$ with $|K| \leq k$, the difference between their projected energies satisfies $|W(f_K) - W(g_K)| \leq 3\epsilon$.

While Theorem 1 guarantees that the hypothesis g returned by the IKM algorithm is sparse, this sparsity is strictly "spectral"; that is, $L_0(g)$ is small. The coordinates in $\mathrm{supp}(g)$ might collectively involve a large number of distinct variables, which would still require an exhaustive search over $\binom{d}{k}$ possible subsets to find the optimal K. To achieve fixed-parameter tractability, it is essential to prune the feature space to a compact candidate set I of variables that drive the model's behavior. To this end, we define the *restricted influence* of a feature i of degree k, which aggregates the energy over all coordinates involving i up to support size k:

$$\mathrm{Inf}_i^k[g] = W_{\mathcal{R}_i^k}(g) = \sum_{\boldsymbol{s} \in \mathcal{R}_i^k} \hat{g}(\boldsymbol{s})^2 \quad \text{where} \quad \mathcal{R}_i^k = \{\boldsymbol{s} \in \mathcal{S} : s_i \neq 0, |\mathrm{supp}(\boldsymbol{s})| \leq k\}.$$

The following lemma establishes that features with low restricted influence can be safely disregarded.

Lemma 6. For a function $g : \mathbf{\Omega} \to \mathbb{R}$ and $\epsilon > 0$, let $I = \{i \in [d] : \mathrm{Inf}_i^k[g] \geq \epsilon/k\}$ be the set of influential features. Then for any $K \subseteq [d]$ with $|K| \leq k$, it holds that $W(g_K) \leq W(g_{K \cap I}) + \epsilon$.

Based on these preparatory results, Algorithm 2 implements the spectral approach to global explainability: it first applies the IKM algorithm to obtain a spectral hypothesis g that approximates f, then prunes the feature space to a candidate set I of influential variables, and finally performs an exhaustive search within this reduced set. Both the influences and the projected energies are computed directly from the coefficients of g, since $W(g_K) = \sum\{\hat{g}(\boldsymbol{s})^2 : \boldsymbol{s} \in \mathrm{supp}(g),\ \mathrm{supp}(\boldsymbol{s}) \subseteq K\}$. The following theorem establishes that this algorithm is fixed-parameter tractable in k, c, and $1/\epsilon$, and achieves near-optimal relevance with high probability.

Theorem 2. There exists an algorithm such that, for any d-dimensional feature space $\mathbf{\Omega}$ with maximum alphabet size c, any binary classifier f over $\mathbf{\Omega}$, and any product distribution $\mathcal{D}$ with full support over $\mathbf{\Omega}$, given any size constraint $k \leq d$, parameters $\epsilon, \delta \in (0, 1)$, and membership query access to f, the algorithm returns a subset $K \subseteq [d]$ of size at most k satisfying, with probability at least $1 - \delta$:

$$\mathsf{err}_{f,\mathcal{D}}(K) \leq \min_{K' \subseteq [d], |K'| \leq k} \mathsf{err}_{f,\mathcal{D}}(K') + \epsilon.$$

The algorithm is fixed-parameter tractable in k, c, and $1/\epsilon$, with a runtime polynomial in d, $(ck/\epsilon)^k$, and $\ln(1/\delta)$.

Sketch of proof. Let K be returned by Algorithm 2, and K^* be optimal. By Lemma 4, it suffices to show that $W(f_{K^*}) - W(f_K) \leq 2\epsilon$. With probability at least $1-\delta$, IKM returns g with $L_\infty(f-g) \leq \theta$ (Theorem 1). Then, Lemma 5 changes the energy of K^* and of K by at most $3\epsilon'$ each when passing from f to g, Lemma 6 loses at most ϵ' when restricting K^* to I, and K maximizes $W(g_{K'})$ over all subsets $K' \subseteq I$ of size at most k: hence $W(f_{K^*}) - W(f_K) \leq 7\epsilon' = 2\epsilon$.

For the runtime, Theorem 1 with $\theta = 2\epsilon/(7c^{k/2})$ shows that IKM is polynomial in d, c^k/ϵ, and $\ln(1/\delta)$; the key point is the size of I. By soundness, every $\boldsymbol{s} \in \mathrm{supp}(g)$ satisfies $|\hat{f}(\boldsymbol{s})| \geq \theta/2$, and its estimate is accurate, so $|\hat{g}(\boldsymbol{s})| \leq |\hat{f}(\boldsymbol{s})| + \theta^2/4 \leq \frac{3}{2}|\hat{f}(\boldsymbol{s})|$. By Parseval's identity, it follows that $W(g) \leq 9/4$. Since a coordinate of support size at most k contributes to at most k restricted influences, $\sum_i \mathrm{Inf}_i^k[g] \leq 9k/4$, so at most $|I| \leq (9k/4)/(\epsilon'/k) = 63k^2/(8\epsilon)$ features pass the threshold ϵ'/k. The exhaustive search therefore examines at most $(63ek/(8\epsilon))^k$ subsets, each evaluated from the $L_0(g) = O(c^k/\epsilon^2)$ coefficients of g. □

5 Experimental Evaluation

Our experiments address three questions. (Q1) Does the Spectral Explainer (SE, Algorithm 2) return explanations within ϵ of the optimum, with a failure rate at most δ, as guaranteed by Theorem 2? (Q2) Does its query complexity scale with d, k and c as predicted by the analysis? (Q3) How does it compare to the Greedy Explanation baseline (GE) inspired by [6] on real classifiers, and are the returned explanations meaningful? Both algorithms are implemented in `Python 3.14` with `PyTorch`, and were run on a workstation with an Intel Xeon Gold 6252 CPU, 384 GB of RAM, and two NVIDIA Quadro RTX 8000 GPUs. The detailed protocol and complete results are in Appendix B.

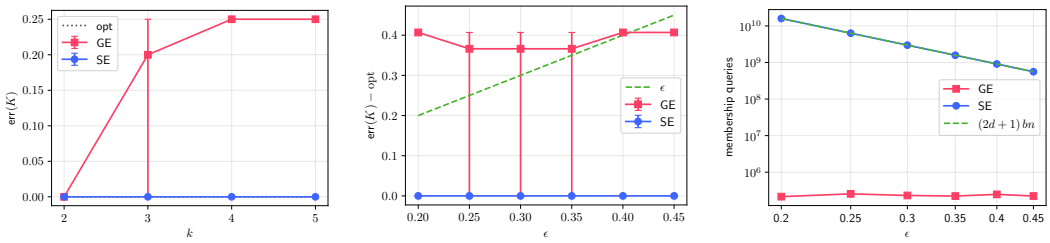
Explainers. SE follows Algorithm 2: given (ϵ, δ, k), it sets $\epsilon' = 2\epsilon/7$ and $\theta = \epsilon'/c^{k/2}$, runs IKM with confidence δ_0 and b blocks as in Algorithm 1, and prunes features at ϵ'/k. The only deviation from the analysis is the block size $n = \lceil \kappa \cdot 64/\theta^4 \rceil$. While Lemma 3 requires $\kappa = 1$, which is conservative, calibration on hidden parities showed that $\kappa = 0.003$ was the smallest value with no failures. We use $\kappa = 0.01$, that is, 100 times fewer samples than prescribed. Our experiments thus test a stronger claim than Theorem 2: if the failure rate stays within δ under this reduced budget, the guarantee holds *a fortiori* for $\kappa = 1$, which only makes estimates more accurate. Three implementation choices leave the analysis unchanged (Appendix B.1): IKM estimates all prefixes of a given depth on a shared batch (Remark A.1), and discards coordinates of degree greater than k, which are never used afterwards; moreover, since $\mathsf{err}_{f,\mathcal{D}}$ is non-increasing under inclusion, SE completes its explanation up to k features, by decreasing order of estimated influence, whenever fewer than k features pass the pruning. The baseline GE performs forward selection: it starts from $K = \varnothing$ and, at each step, adds the feature i that minimizes $\mathsf{err}_{f,\mathcal{D}}(K \cup \{i\})$, estimated on $5{,}000$ fresh instances per candidate, until $|K| = k$ or no feature further reduces the error.

Evaluation. Each explanation K is scored by its exact error $\mathsf{err}_{f,\mathcal{D}}(K) = (1 - W(f_K))/2$, computed by enumerating $\boldsymbol{\Omega}$,[1] and compared to $\mathrm{opt} = \min_{|K'| \leq k} \mathsf{err}_{f,\mathcal{D}}(K')$, computed in closed form or by brute force. A run is a *success* if $\mathsf{err}_{f,\mathcal{D}}(K) \leq \mathrm{opt} + \epsilon$. Unless otherwise specified, $\epsilon = 0.35$ and $\delta = 0.1$. Each synthetic setting is repeated 10 times (100 times for the δ-sweep), and each real-world setting 6 times for every $k \in \{1, \ldots, 5\}$.

Synthetic targets. All targets are defined on $\{0, \ldots, c-1\}^d$ under the uniform distribution, and built from the hidden parity $\chi_S(\boldsymbol{x}) = \prod_{i \in S}(-1)^{x_i}$. For even c, the whole energy of χ_S lies on coordinates with support S, so that $\mathsf{err}_{f,\mathcal{D}}(K) = 0$ if $S \subseteq K$, and $1/2$ otherwise: no proper subset of S explains χ_S better than $\varnothing$, and forward selection receives no signal. We consider three targets: (i) the hidden parity χ_S itself, with $|S| = k$; (ii) a *noisy parity*, where the labels of χ_S ($|S| = 3$) are flipped on a fixed pseudo-random subset of density $\eta = 0.05$, so that $\mathrm{opt} \approx 2\eta(1-\eta) \approx 0.093$, attained by $K = S$; (iii) the *Synergy Trap*, defined by $f(\boldsymbol{x}) = 1$ if $x_{i^*} = 1$ and $f(\boldsymbol{x}) = \chi_S(\boldsymbol{x})$ otherwise, with $i^* \notin S$ and $|S| = k - 1$. In the Synergy Trap, $\mathsf{err}_{f,\mathcal{D}}(\varnothing) = 3/8$ and $\mathsf{err}_{f,\mathcal{D}}(\{i^*\}) = 1/4$, while adding any proper subset of S to $\{i^*\}$ leaves the error unchanged: the distractor i^* is the only feature that helps on its own, yet only $\{i^*\} \cup S$ reaches $\mathsf{err}_{f,\mathcal{D}}(K) = 0$.

Real-world benchmarks. We used nine binary classification tasks from `OpenML` with d ranging from 8 to 20, which allows the exact computation of opt. Instances with a missing label were removed, as were attributes with any missing value, and the remaining attributes were binarized ($c = 2$) using a median split for numerical variables and a most-frequent-value test for categorical

[1] When $|\boldsymbol{\Omega}| > 2^{22}$, $\mathsf{err}_{f,\mathcal{D}}(K)$ is estimated on 10^6 instances.



(a) Synergy Trap (b) Noisy parity: gap (c) Noisy parity: queries

Figure 1: Synthetic targets ($d = 16$); bars show the range over runs. (a) $\mathsf{err}_{f,\mathcal{D}}(K)$ against k, with $\mathrm{opt} = 0$. (b) $\mathsf{err}_{f,\mathcal{D}}(K) - \mathrm{opt}$ against ϵ, with the bound ϵ (dashed). (c) Membership queries against ϵ, with the value $(2d+1)\,b\,n$ of the analysis (dashed).

Table 1: Growth of the number of queries $Q = (2d+1)\,b\,n$ of SE along three sweeps on hidden parities ($\epsilon = 0.35$, $\delta = 0.1$): the observed factor is the product of the three predicted ones.

Sweep	$2d+1$	$n \propto c^{2k}$	b	Q (observed)
d: $8 \to 128$ ($k = 3$, $c = 2$)	$\times 15.1$	$\times 1$	$\times 1.20$	$\times 18.1$
k: $k \to k+1$ ($d = 16$, $c = 2$)	$\times 1$	$\times 4$	$\times 1.04$ to $\times 1.06$	$\times 4.2$
c: $2 \to 6$ ($d = 16$, $k = 2$)	$\times 1$	$\times 81$	$\times 1.24$	$\times 101$

ones. For each dataset, f is a multi-layer perceptron with one hidden layer of 100 ReLU units, trained on 80% of the data, and $\mathcal{D}$ is the product of empirical marginals with Laplace smoothing.

Results on synthetic targets (Q1, Q2). *Accuracy.* The Synergy Trap ($d = 16$, Figure 1a) clearly separates the two explainers: SE returns $\{i^*\} \cup S$ in all 40 runs ($\mathsf{err}_{f,\mathcal{D}}(K) = 0$), while GE selects the distractor i^* first and then stops, since no single feature of S improves the error; it ends with $\mathsf{err}_{f,\mathcal{D}}(K) = 1/4$ in 28 of the 30 runs with $k \geq 3$. On the noisy parity ($d = 16$, $k = 3$, Figure 1b), the gap $\mathsf{err}_{f,\mathcal{D}}(K) - \mathrm{opt}$ of SE is zero for all $\epsilon \in [0.20, 0.45]$, whereas GE misses the parity in most runs, with an average gap close to 0.4. On hidden parities, SE reached opt in all 130 runs of the cost sweeps below, against 7 of the 120 runs with $|S| \geq 2$ for GE.

Confidence. On the noisy parity, SE did not fail in any of 600 runs for six values of δ ranging from 0.01 to 0.5 (Figure 2 in Appendix B.3): the observed failure rate remains far below δ, consistent with the conservative bound of Lemma 3.

Cost. In every run, the number of queries of SE is exactly $Q = (2d+1)\,b\,n$ (Figure 1c), as expected with shared batches. Table 1 decomposes the growth of Q along three sweeps on hidden parities into the three factors of this product: Q is linear in d up to the logarithmic factor b, and grows as $n \propto \theta^{-4} = c^{2k}/\epsilon'^4$ in k and c. The search itself remains cheap: IKM performs between 18 and 251 estimations on average, several orders of magnitude below its bound $8dc/\theta^2$, because the pruning at $\theta^2/2$ discards almost every prefix of a parity (Figure 3 in Appendix B.3).

Results on real-world benchmarks (Q3). Table 2 reports five of the nine datasets, as the remaining four yield similar conclusions. SE achieves the exact optimum in 138 out of 150 runs, compared to 110 for GE (246 and 211 out of 270 across all datasets). Every run of both explainers satisfies $\mathsf{err}_{f,\mathcal{D}}(K) \leq \mathrm{opt} + \epsilon$ by a wide margin: the largest gap is 0.004 for SE and 0.018 for GE. The few misses by SE are near-ties between explanations that the estimated spectrum does not resolve. GE misses more frequently, notably on Tic-Tac-Toe, where the label is a logical function with strong feature interactions. SE is also more stable: it returned the same explanation in all runs for each of the 45 (dataset, k) pairs, compared to only 24 for GE. This accuracy comes at a cost: since n grows as θ^{-4}, a run of SE at $k = 5$ takes between 4 and 41 minutes on these datasets, while GE completes in a fraction of a second.

Explanations. Table 3 illustrates the explanations returned by SE for $k = 3$. Recall that a "global" explanation is a set of features, not a conjunction of literals: it states which tests suffice to determine the prediction of f on average, whatever their outcomes. The resulting sets are directly interpretable.

Table 2: Real-world benchmarks ($\epsilon = 0.35$, $\delta = 0.1$). "opt reached": runs with $\mathsf{err}_{f,\mathcal{D}}(K) = \mathrm{opt}$, over $k \in \{1, \ldots, 5\}$ and 6 runs. SE returned the same explanation in all runs, hence a single value. Time: one run of SE at $k = 5$. Acc.: test accuracy of f. All nine datasets are reported in Appendix B.

			opt reached		$\mathsf{err}_{f,\mathcal{D}}(K)$ at $k = 5$			
Dataset	d	Acc.	SE	GE	SE	GE	opt	Time (s)
Breast Cancer (WI)	8	0.98	30/30	25/30	0.080	0.080 ± 0.000	0.080	244
Tic-Tac-Toe	9	0.94	30/30	21/30	0.249	0.254 ± 0.007	0.249	397
COMPAS	13	0.62	30/30	26/30	0.110	0.112 ± 0.003	0.110	949
Mushroom	20	1.00	24/30	17/30	0.183	0.185 ± 0.003	0.183	1998
Credit-G	20	0.70	24/30	21/30	0.217	0.221 ± 0.004	0.217	2464

Table 3: Explanations returned by SE at $k = 3$, all optimal and identical across runs. Each feature is denoted by the binary test that defines it; the explanation partitions $\mathbf{\Omega}$ into the $2^3 = 8$ regions defined by the outcomes of its three tests. The other four datasets are in Appendix B.4.

Dataset	Explanation K returned by SE ($k = 3$)	$\mathsf{err}_{f,\mathcal{D}}(K)$
Breast Cancer (WI)	clump thickness > 4, cell shape uniformity > 1, bland chromatin > 3	0.139
Tic-Tac-Toe	top-left = x, middle-middle = x, bottom-left = x	0.326
COMPAS	age > 31, priors count > 2, age category = less than 25	0.138
Mushroom	odor = none, gill size = broad, stalk surface above ring = smooth	0.235
Credit-G	checking status = no checking, duration > 18, savings < 100	0.247

For example, in Tic-Tac-Toe, f is best explained by the center and two corner squares, which lie on four and three winning lines respectively, more than any other square. This triple also reveals feature interactions: greedy selection always picks the center first, but then selects the wrong corner in three out of six runs at $k = 2$, and misses the optimal triple once at $k = 3$. In COMPAS, predictions depend mainly on age and the number of prior offenses; in Credit-G, the key features are checking account status, loan duration, and savings.

6 Limitations and Perspectives

We introduced a spectral framework for global explainability as a principled alternative to greedy selection. Using an integer extension of the KM algorithm, we established PAC-FPT guarantees for learning global explanations in discrete feature spaces. Experiments confirm these guarantees: SE recovers synergistic interactions missed by forward selection, reaches the exact optimum on most real-world benchmarks, and consistently returns the same explanation across runs.

Our study also reveals two main limitations. First, although SE is fixed-parameter tractable, its sample size grows as $\theta^{-4} \propto c^{2k}/\epsilon^4$ (Theorem 1): a run takes tens of minutes for $d = 20$ and $k = 5$, and several hours for $d \approx 60$. In the latter regime, the energy of f is spread over many features, so that none passes pruning and the guarantee holds trivially, as ϵ exceeds the gain of any explanation over $\varnothing$. Still, our bounds leave room for improvement, since 300 times fewer samples sufficed in calibration. Future work will explore tighter, adaptive spectral estimates, solver-inspired pruning, and faster prefix-tree traversal heuristics.

Second, our framework assumes a product distribution $\mathcal{D}$ over a discrete space $\mathbf{\Omega}$. While this assumption is reasonable for many tabular tasks, it does not capture the complex dependencies present in real-world data. Extending global explainability to distributions factorized by generative models—such as Bayesian networks or probabilistic circuits—is a crucial direction for future research. In this regard, Heidari and Khardon [18] recently extended the KM algorithm to Bayesian network distributions over Boolean domains to demonstrate that DNF formulas are agnostically PAC learnable. However, learning global probabilistic explanations in this broader context remains an open challenge. Generalizing such spectral techniques to factorized distributions over integer-valued domains could yield explanations that are more closely aligned with the true joint structure of the data.

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A Technical Appendix: Proofs and Algorithmic Details

This section provides the proofs of the main results stated in the paper, along with additional technical lemmas and details of the algorithms.

A.1 Fourier Bases

An orthonormal basis for $\mathcal{L}^2(\Omega, \mathcal{D})$ can be constructed using the Stieltjes procedure (Algorithm 3), which employs a three-term recurrence to generate orthonormal polynomials, where ϕ_j has degree j. For ordinal features, this hierarchy is semantically rich: $\phi_0 = 1$ represents the constant component (the mean), ϕ_1 captures linear trends, and ϕ_2 captures quadratic curvature. For categorical features, the values $0, \ldots, c-1$ simply serve as distinct nodes, and the procedure still yields a valid basis.

Algorithm 3: Stieltjes Procedure

Input: Distribution $\mathcal{D}$ over $\Omega = \{0, 1, \ldots, c-1\}$
Output: Orthonormal basis functions $\{\phi_0, \phi_1, \ldots, \phi_{c-1}\}$

Initialize $\phi_{-1}(x) \leftarrow 0$, $\phi_0(x) \leftarrow 1$, and $\alpha_0 \leftarrow 0$
for $j = 0$ **to** $c-2$ **do**
  $\beta_j \leftarrow \sum_{a=0}^{c-1} \mathcal{D}(a) \cdot a \cdot \phi_j(a)^2$ // $\beta_j = \langle x\phi_j, \phi_j \rangle_{\mathcal{D}}$
  $u_{j+1}(x) \leftarrow (x - \beta_j)\phi_j(x) - \alpha_j \phi_{j-1}(x)$ // Orthogonalization step
  $\alpha_{j+1} \leftarrow \sqrt{\sum_{a=0}^{c-1} \mathcal{D}(a) \cdot u_{j+1}(a)^2}$ // Normalization
  $\phi_{j+1}(x) \leftarrow u_{j+1}(x)/\alpha_{j+1}$
return $\{\phi_0, \phi_1, \ldots, \phi_{c-1}\}$

Lemma A.1. For any distribution $\mathcal{D}$ with full support over an alphabet Ω of size c, an orthonormal basis $\{\phi_0, \ldots, \phi_{c-1}\}$ for $\mathcal{L}^2(\Omega, \mathcal{D})$ with $\phi_0 = 1$ can be computed in $O(c^2)$ time using the Stieltjes procedure [15].

Proof. We show by induction on j that $\phi_0, \ldots, \phi_j$ are orthonormal and that each ϕ_k is a polynomial of degree exactly k. The base case $j = 0$ holds since $\langle \phi_0, \phi_0 \rangle_{\mathcal{D}} = \sum_a \mathcal{D}(a) = 1$. For the inductive step, note first that $\phi_0, \ldots, \phi_j$ span all polynomials of degree at most j, so ϕ_j is orthogonal to every polynomial of degree less than j.

Well-definedness. By construction, u_{j+1} is a polynomial of degree exactly $j+1 \le c-1$. Since $\mathcal{D}$ has full support, $\alpha_{j+1}^2 = \sum_a \mathcal{D}(a)\, u_{j+1}(a)^2 = 0$ would force u_{j+1} to vanish on all c points of Ω, which is impossible for a nonzero polynomial of degree at most $c-1$. Hence $\alpha_{j+1} > 0$, and ϕ_{j+1} has unit norm and degree $j+1$.

Orthogonality. For $k \le j$, using $\langle x\phi_j, \phi_k \rangle_{\mathcal{D}} = \langle \phi_j, x\phi_k \rangle_{\mathcal{D}}$, we get

$$\langle u_{j+1}, \phi_k \rangle_{\mathcal{D}} = \langle \phi_j, x\phi_k \rangle_{\mathcal{D}} - \beta_j \mathbb{I}[k = j] - \alpha_j \mathbb{I}[k = j-1].$$

If $k < j-1$, the polynomial $x\phi_k$ has degree $k+1 < j$, so all three terms vanish. If $k = j$, the first term equals β_j by definition. If $k = j-1$, the recurrence at step $j-1$ gives $x\phi_{j-1} = \alpha_j \phi_j + \beta_{j-1}\phi_{j-1} + \alpha_{j-1}\phi_{j-2}$, so the first term equals α_j. In all cases, $\langle u_{j+1}, \phi_k \rangle_{\mathcal{D}} = 0$. Finally, c orthonormal functions in the c-dimensional space $\mathcal{L}^2(\Omega, \mathcal{D})$ form a basis.

Regarding runtime, each function is represented by its c values on Ω, so each of the $c-1$ iterations takes $O(c)$ time, for a total of $O(c^2)$. □

For each feature $i \in [d]$, let $\{\phi_{i,0}, \ldots, \phi_{i,c_i-1}\}$ be the orthonormal basis of $\mathcal{L}^2(\Omega_i, \mathcal{D}_i)$ given by Lemma A.1. By taking the tensor product of these local bases, we derive a global Fourier basis for the product space $\boldsymbol{\Omega} = \Omega_1 \times \cdots \times \Omega_d$. Under the product distribution $\mathcal{D}$, this basis enables a unique representation of any function $f : \boldsymbol{\Omega} \to \mathbb{R}$ as a linear combination of basis functions. Let $\mathcal{S} = \prod_{i=1}^{d} \{0, 1, \ldots, c_i - 1\}$ denote the set of multi-indices, referred to as *coordinates*. The *global basis function* of a multi-index $\boldsymbol{s} \in \mathcal{S}$ is defined as $\phi_{\boldsymbol{s}}(\boldsymbol{x}) = \prod_{i=1}^{d} \phi_{i,s_i}(x_i)$.

Lemma A.2. Let $\mathcal{D}$ be a product distribution with full support over $\mathbf{\Omega}$. The collection $\{\phi_{\boldsymbol{s}} : \boldsymbol{s} \in \mathcal{S}\}$ forms a complete orthonormal basis for $\mathcal{L}^2(\mathbf{\Omega}, \mathcal{D})$.

Proof. For any pair $\boldsymbol{s}, \boldsymbol{t} \in \mathcal{S}$, the product structure of $\mathcal{D}$ and the definition of $\phi_{\boldsymbol{s}}$ yield:

$$\langle \phi_{\boldsymbol{s}}, \phi_{\boldsymbol{t}} \rangle_{\mathcal{D}} = \prod_{i=1}^{d} \mathbb{E}_{x_i \sim \mathcal{D}_i}[\phi_{i,s_i}(x_i)\phi_{i,t_i}(x_i)] = \prod_{i=1}^{d} \mathbb{I}[s_i = t_i] = \mathbb{I}[\boldsymbol{s} = \boldsymbol{t}],$$

following from the orthonormality of local bases (Lemma A.1). Since $\mathcal{D}$ has full support, the dimension of $\mathcal{L}^2(\mathbf{\Omega}, \mathcal{D})$ is $|\mathbf{\Omega}| = \prod_i c_i = |\mathcal{S}|$, so this orthonormal set is complete. □

Corollary A.1. Let $f, g \in \mathcal{L}^2(\mathbf{\Omega}, \mathcal{D})$. Then, $\langle f, g \rangle_{\mathcal{D}} = \sum_{\boldsymbol{s} \in \mathcal{S}} \hat{f}(\boldsymbol{s})\hat{g}(\boldsymbol{s})$. Furthermore, $\|f\|_2^2 = L_2(f)^2$, which equals 1 if f takes values in $\{-1, +1\}$.

Proof. The first equality (Plancherel's theorem) follows directly from the orthonormality of $\{\phi_{\boldsymbol{s}}\}_{\boldsymbol{s} \in \mathcal{S}}$ established in Lemma A.2. The second equality (Parseval's identity) is the case $f = g$. In particular, if f is a classifier taking values in $\{-1, +1\}$, then $\|f\|_2^2 = \mathbb{E}[f(\boldsymbol{x})^2] = 1$, so $L_2(f)^2 = 1$. □

A.2 The Integer Kushilevitz–Mansour Algorithm

To demonstrate both the efficiency and correctness of the IKM algorithm, we begin by presenting a concentration bound for the MoM estimator. This estimator is particularly valuable for product distributions, where local basis functions can have large ranges, as it relies on variance instead of boundedness. Although this result was already established in [28], we provide a self-contained proof for completeness.

Proposition A.1. Let X be a random variable with mean μ and variance at most $\sigma^2 < \infty$, and let $t > 0$ and $\delta_0 \in (0, 1)$. Given $b \geq 8 \ln(1/\delta_0)$ blocks of $n \geq 4\sigma^2/t^2$ i.i.d. copies of X each, the MoM estimator $\hat{\mu} = \mathrm{median}(\bar{X}_1, \ldots, \bar{X}_b)$ satisfies $|\hat{\mu} - \mu| \leq t$ with probability at least $1 - \delta_0$, where $\bar{X}_j$ is the empirical mean of the j-th block.

Proof. Each block mean satisfies $\mathbb{E}[\bar{X}_j] = \mu$ and $\mathrm{Var}[\bar{X}_j] \leq \sigma^2/n$. By Chebyshev's inequality and the choice of n,

$$\mathbb{P}(|\bar{X}_j - \mu| > t) \leq \frac{\sigma^2}{nt^2} \leq \frac{1}{4}.$$

Let $E_j = \mathbb{I}[|\bar{X}_j - \mu| > t]$ and $B = \sum_{j=1}^{b} E_j$. The variables E_j are independent, since the blocks are disjoint, and $\mathbb{E}[B] \leq b/4$. If $|\hat{\mu} - \mu| > t$, then at least half of the block means lie on the same side of μ at distance more than t, so $B \geq b/2$. By Hoeffding's inequality,

$$\mathbb{P}(|\hat{\mu} - \mu| > t) \leq \mathbb{P}\left(B - \mathbb{E}[B] \geq \frac{b}{4}\right) \leq e^{-2b(1/4)^2} = e^{-b/8} \leq \delta_0,$$

where the last inequality follows from the choice of b. □

Applying this result to a classifier f taking values in $\{-1, +1\}$, we note that the random variables averaged by our estimators have variance at most 1, independently of the marginal probabilities of the distribution. This yields the following guarantees, which restate Lemma 3.

Lemma A.3. Let $f : \mathbf{\Omega} \to \{-1, +1\}$ be a classifier and $\mathcal{D}$ a product distribution with full support. For any prefix $\boldsymbol{\sigma}$ and coordinate $\boldsymbol{s}$, the random variables averaged by WEIGHT and COEFFICIENT have expectations $W_{\boldsymbol{\sigma}}$ and $\hat{f}(\boldsymbol{s})$, respectively, and variance at most 1. Consequently, with $b \geq 8 \ln(1/\delta_0)$ blocks of $n \geq 64/\theta^4$ samples each, every call to WEIGHT or COEFFICIENT returns an estimate within $\theta^2/4$ of its target with probability at least $1 - \delta_0$.

Proof. Once the expectations and the variance bound are established, the result follows from Proposition A.1 with $\sigma^2 = 1$ and $t = \theta^2/4$, since then $4\sigma^2/t^2 = 64/\theta^4$.

Coefficient estimation. The procedure COEFFICIENT averages $X = f(\boldsymbol{z})\phi_{\boldsymbol{s}}(\boldsymbol{z})$, where $\boldsymbol{z} \sim \mathcal{D}$. Its expectation is $\langle f, \phi_{\boldsymbol{s}} \rangle_{\mathcal{D}} = \hat{f}(\boldsymbol{s})$. Since $f^2 = 1$ and $\mathbb{E}[\phi_{\boldsymbol{s}}^2] = 1$, we have $\mathbb{E}[X^2] = 1$, hence $\mathrm{Var}[X] = 1 - \hat{f}(\boldsymbol{s})^2 \leq 1$.

Weight estimation. Let p be the length of $\boldsymbol{\sigma}$. The procedure WEIGHT averages $X = f(\boldsymbol{u}, \boldsymbol{v})\, \phi_{\boldsymbol{\sigma}}(\boldsymbol{u})\, f(\boldsymbol{u}', \boldsymbol{v})\, \phi_{\boldsymbol{\sigma}}(\boldsymbol{u}')$, where $\boldsymbol{u}, \boldsymbol{u}' \sim \mathcal{D}_{1:p}$ and $\boldsymbol{v} \sim \mathcal{D}_{p+1:d}$ are independent. Conditioning on $\boldsymbol{v}$ and using the independence of $\boldsymbol{u}$ and $\boldsymbol{u}'$,

$$\mathbb{E}[X \mid \boldsymbol{v}] = h_{\boldsymbol{\sigma}}(\boldsymbol{v})^2, \quad \text{where} \quad h_{\boldsymbol{\sigma}}(\boldsymbol{v}) = \mathbb{E}_{\boldsymbol{u}}[f(\boldsymbol{u}, \boldsymbol{v})\, \phi_{\boldsymbol{\sigma}}(\boldsymbol{u})].$$

Expanding f in the global basis and using the orthonormality of $\{\phi_{\boldsymbol{\sigma}'}\}$ over prefixes of length p, we get $h_{\boldsymbol{\sigma}}(\boldsymbol{v}) = \sum_{\boldsymbol{s} \in \mathcal{C}_{\boldsymbol{\sigma}}} \hat{f}(\boldsymbol{s})\, \phi_{\boldsymbol{s}_{p+1:d}}(\boldsymbol{v})$, where $\phi_{\boldsymbol{s}_{p+1:d}}(\boldsymbol{v}) = \prod_{i=p+1}^{d} \phi_{i,s_i}(v_i)$. These suffix basis functions are orthonormal under $\mathcal{D}_{p+1:d}$, so

$$\mathbb{E}[X] = \mathbb{E}_{\boldsymbol{v}}[h_{\boldsymbol{\sigma}}(\boldsymbol{v})^2] = \sum_{\boldsymbol{s} \in \mathcal{C}_{\boldsymbol{\sigma}}} \hat{f}(\boldsymbol{s})^2 = W_{\boldsymbol{\sigma}}.$$

For the variance, using $f^2 = 1$ and the independence of $\boldsymbol{u}$ and $\boldsymbol{u}'$,

$$\mathbb{E}[X^2] = \mathbb{E}_{\boldsymbol{u}}[\phi_{\boldsymbol{\sigma}}(\boldsymbol{u})^2] \cdot \mathbb{E}_{\boldsymbol{u}'}[\phi_{\boldsymbol{\sigma}}(\boldsymbol{u}')^2] = 1,$$

hence $\mathrm{Var}[X] \leq 1$. □

The logical structure of the search procedure guarantees that the algorithm identifies all significant spectral energy while ensuring the resulting hypothesis remains sparse. In what follows, we assume $c \geq 2$ (a feature with a single value is constant and can be discarded).

Proposition A.2. Let g be the hypothesis produced by the IKM algorithm with threshold $\theta \in (0, 1]$ and confidence $\delta \in (0, 1)$, that is, with $\delta_0 = \theta^2\delta/(8dc)$, $b = \lceil 8 \ln(1/\delta_0) \rceil$, and $n = \lceil 64/\theta^4 \rceil$. Then, with probability at least $1 - \delta$, IKM makes at most $N = 8dc/\theta^2$ calls to WEIGHT and COEFFICIENT, and the following properties hold:

Completeness: For every $\boldsymbol{s}$ such that $|\hat{f}(\boldsymbol{s})| \geq \theta$, we have $\boldsymbol{s} \in \mathrm{supp}(g)$ and $|\hat{g}(\boldsymbol{s}) - \hat{f}(\boldsymbol{s})| \leq \theta^2/4$.

Soundness: For every $\boldsymbol{s} \in \mathrm{supp}(g)$, we have $|\hat{f}(\boldsymbol{s})| \geq \theta/2$.

Proof. Call an estimate *accurate* if it lies within $\theta^2/4$ of its target, and let G be the event that the first $\min(N, T)$ calls are accurate, where T is the (random) total number of calls. Each call uses fresh samples, so conditionally on the past, it is accurate with probability at least $1 - \delta_0$ by Lemma A.3. By a union bound over at most N calls, $\mathbb{P}(G) \geq 1 - N\delta_0 = 1 - \delta$. We now show that, on G, the algorithm makes at most N calls, so that all its estimates are accurate, and that both properties hold.

Counting. On G, consider the calls made so far. A prefix $\boldsymbol{\sigma}$ is retained only if $\hat{W}_{\boldsymbol{\sigma}} \geq \theta^2/2$, which implies $W_{\boldsymbol{\sigma}} \geq \theta^2/4$. Prefix cones of a given length p partition $\mathcal{S}$, so by Parseval's identity, their weights sum to $\|f\|_2^2 = 1$. Hence at most $4/\theta^2$ prefixes are retained at each depth. Each retained prefix of length $p < d$ spawns at most c children, so WEIGHT is called on at most $4c/\theta^2$ prefixes at each depth $p \geq 1$, plus once on $\boldsymbol{\lambda}$, and COEFFICIENT is called on at most $4/\theta^2$ leaves. The total number of calls is thus at most

$$1 + \frac{4dc}{\theta^2} + \frac{4}{\theta^2} \leq \frac{8dc}{\theta^2} = N,$$

where the inequality uses $dc \geq 2$ and $\theta \leq 1$.

Completeness. Suppose that $|\hat{f}(\boldsymbol{s})| \geq \theta$. For any prefix $\boldsymbol{\sigma}$ of $\boldsymbol{s}$, we have $\boldsymbol{s} \in \mathcal{C}_{\boldsymbol{\sigma}}$, so $W_{\boldsymbol{\sigma}} \geq \hat{f}(\boldsymbol{s})^2 \geq \theta^2$, and therefore

$$\hat{W}_{\boldsymbol{\sigma}} \geq W_{\boldsymbol{\sigma}} - \frac{\theta^2}{4} \geq \frac{3\theta^2}{4} > \frac{\theta^2}{2}.$$

Hence the path to $\boldsymbol{s}$ is never pruned, and $\boldsymbol{s}$ is added to g with $|\hat{g}(\boldsymbol{s}) - \hat{f}(\boldsymbol{s})| \leq \theta^2/4$.

Soundness. Suppose $\boldsymbol{s} \in \mathrm{supp}(g)$. At depth d, the cone of $\boldsymbol{s}$ is $\{\boldsymbol{s}\}$, so $W_{\boldsymbol{s}} = \hat{f}(\boldsymbol{s})^2$. Since $\boldsymbol{s}$ was retained, $\hat{W}_{\boldsymbol{s}} \geq \theta^2/2$, and therefore $\hat{f}(\boldsymbol{s})^2 \geq \theta^2/2 - \theta^2/4 = \theta^2/4$, that is, $|\hat{f}(\boldsymbol{s})| \geq \theta/2$. □

Theorem A.1. Let $f : \mathbf{\Omega} \to \{-1, +1\}$ be a classifier and $\mathcal{D}$ a product distribution with full support and maximum alphabet size c. Assume that each membership query to f and each draw from a marginal $\mathcal{D}_i$ take $O(1)$ time. Then, for any threshold $\theta \in (0, 1]$ and confidence level $\delta \in (0, 1)$, the IKM algorithm makes $O\big(\frac{dc}{\theta^6} \ln \frac{dc}{\theta\delta}\big)$ membership queries, runs in time $O\big(dc^2 + \frac{d^2c}{\theta^6} \ln \frac{dc}{\theta\delta}\big)$, and returns a hypothesis g such that, with probability at least $1 - \delta$:

$$L_\infty(f - g) \le \theta \quad \text{and} \quad L_0(g) \le \frac{4}{\theta^2}.$$

Proof. We work on the event of Proposition A.2, which has probability at least $1 - \delta$. By soundness, every $\boldsymbol{s} \in \mathrm{supp}(g)$ satisfies $\hat{f}(\boldsymbol{s})^2 \ge \theta^2/4$, so by Parseval's identity,

$$1 = \sum_{\boldsymbol{s} \in \mathcal{S}} \hat{f}(\boldsymbol{s})^2 \ge \sum_{\boldsymbol{s} \in \mathrm{supp}(g)} \hat{f}(\boldsymbol{s})^2 \ge L_0(g) \cdot \frac{\theta^2}{4},$$

which yields $L_0(g) \le 4/\theta^2$. For $\boldsymbol{s} \in \mathrm{supp}(g)$, the estimate is accurate, so $|\hat{g}(\boldsymbol{s}) - \hat{f}(\boldsymbol{s})| \le \theta^2/4 \le \theta$. For $\boldsymbol{s} \notin \mathrm{supp}(g)$, completeness implies $|\hat{f}(\boldsymbol{s})| < \theta$, and $\hat{g}(\boldsymbol{s}) = 0$. Thus, $L_\infty(f - g) \le \theta$.

For the complexity, IKM makes at most $N = O(dc/\theta^2)$ calls, each using $bn = O(\theta^{-4} \ln(1/\delta_0))$ samples, with at most two membership queries per sample. Since $\ln(1/\delta_0) = \ln \frac{8dc}{\theta^2\delta} = O\big(\ln \frac{dc}{\theta\delta}\big)$, the number of queries is $O\big(\frac{dc}{\theta^6} \ln \frac{dc}{\theta\delta}\big)$. Each sample also requires drawing d coordinates and evaluating a basis function, which is a product of at most d local terms, in $O(d)$ time. Adding the $O(\sum_i c_i^2) = O(dc^2)$ preprocessing time of the Stieltjes procedure (Lemma A.1), the total running time is

$$O\left(dc^2 + \frac{d^2c}{\theta^6} \ln \frac{dc}{\theta\delta}\right).$$

□

Remark A.1 (Shared batches)**.** IKM can process the search depth by depth, estimating at depth p the weights of all the examined prefixes of length p on a single batch of b blocks of n samples, and all the coefficients on a single final batch. The prefixes examined at depth p are determined by the batches of the previous depths, which are independent of the batch of depth p. Hence, conditionally on the past, each estimate is still accurate with probability at least $1 - \delta_0$ by Lemma A.3, and the proof of Proposition A.2 applies verbatim, since its union bound does not require the estimates to be independent. Each batch of depth $1 \le p \le d$ uses two membership queries per sample, and the final batch one, so that IKM makes exactly $(2d + 1)\, b\, n = O\big(\frac{d}{\theta^4} \ln \frac{dc}{\theta\delta}\big)$ queries, independently of the number of examined prefixes. This is the implementation used in Section 5.

A.3 The Spectral Explanation Algorithm

To derive the main result of this paper, we need several preparatory lemmas. The first lemma establishes a structural identity, revealing that the global relevance of any feature subset is precisely quantified by its Fourier energy.

Lemma A.4. For any classifier $f : \mathbf{\Omega} \to \{-1, +1\}$, any product distribution $\mathcal{D}$ with full support on $\mathbf{\Omega}$, and any feature subset K, the global relevance error of K satisfies:

$$\mathsf{err}_{f,\mathcal{D}}(K) = \frac{1}{2}\left(1 - W(f_K)\right).$$

Proof. Let $\boldsymbol{x} \sim \mathcal{D}$ and $\boldsymbol{z} \sim \mathcal{D}(\cdot \mid \boldsymbol{z}_K = \boldsymbol{x}_K)$. Since f takes values in $\{-1, +1\}$, we have $\mathbb{I}[f(\boldsymbol{x}) \ne f(\boldsymbol{z})] = (1 - f(\boldsymbol{x})f(\boldsymbol{z}))/2$, so

$$\mathsf{err}_{f,\mathcal{D}}(K) = \frac{1}{2}\left(1 - \mathbb{E}[f(\boldsymbol{x})f(\boldsymbol{z})]\right).$$

Conditioning on $\boldsymbol{x}$, we get $\mathbb{E}[f(\boldsymbol{z}) \mid \boldsymbol{x}] = f_K(\boldsymbol{x})$ by definition of the projection. Hence, by Plancherel's theorem,

$$\mathbb{E}[f(\boldsymbol{x})f(\boldsymbol{z})] = \mathbb{E}_{\boldsymbol{x} \sim \mathcal{D}}[f(\boldsymbol{x})f_K(\boldsymbol{x})] = \langle f, f_K \rangle_{\mathcal{D}} = \sum_{\boldsymbol{s} \in \mathcal{S}_K} \hat{f}(\boldsymbol{s})^2 = W(f_K),$$

which yields the result. □

The next lemma ensures that the spectral energy of the hypothesis g returned by the IKM algorithm remains a reliable proxy for the energy of f across all possible explanations of size k. For a real-valued function g, the projection g_K is defined as for classifiers, namely $g_K = \sum_{\boldsymbol{s} \in \mathcal{S}_K} \hat{g}(\boldsymbol{s}) \phi_{\boldsymbol{s}}$.

Lemma A.5. Let $f : \mathbf{\Omega} \to \{-1, +1\}$ be a classifier and $g : \mathbf{\Omega} \to \mathbb{R}$ such that $L_\infty(f - g) \leq \epsilon / c^{k/2}$ for some $\epsilon \in (0, 1)$. Then, for any subset $K \subseteq [d]$ of size $|K| \leq k$, the difference between their projected energies satisfies $|W(f_K) - W(g_K)| \leq 3\epsilon$.

Proof. Since $|\mathcal{S}_K| = \prod_{i \in K} c_i \leq c^k$, Parseval's identity gives

$$\|f_K - g_K\|_2^2 = \sum_{\boldsymbol{s} \in \mathcal{S}_K} (\hat{f}(\boldsymbol{s}) - \hat{g}(\boldsymbol{s}))^2 \leq c^k \cdot \frac{\epsilon^2}{c^k} = \epsilon^2.$$

Moreover, $\|f_K\|_2 \leq \|f\|_2 = 1$ and, by the triangle inequality, $\|g_K\|_2 \leq 1 + \epsilon$. Therefore,

$$\begin{aligned} |W(f_K) - W(g_K)| &= \big| \|f_K\|_2 - \|g_K\|_2 \big| \cdot \big(\|f_K\|_2 + \|g_K\|_2 \big) \\ &\leq \|f_K - g_K\|_2 \, (2 + \epsilon) \leq \epsilon(2 + \epsilon) \leq 3\epsilon, \end{aligned}$$

where the last inequality uses $\epsilon < 1$. □

The final lemma establishes that features with low restricted influence can be safely disregarded.

Lemma A.6. For a function $g : \mathbf{\Omega} \to \mathbb{R}$ and $\epsilon > 0$, let $I = \{i \in [d] : \mathrm{Inf}_i^k[g] \geq \epsilon / k\}$ be the set of influential features. Then for any $K \subseteq [d]$ with $|K| \leq k$, it holds that $W(g_K) \leq W(g_{K \cap I}) + \epsilon$.

Proof. Since $\mathcal{S}_{K \cap I} \subseteq \mathcal{S}_K$, we have $W(g_K) - W(g_{K \cap I}) = W_{\mathcal{S}_K \setminus \mathcal{S}_{K \cap I}}(g)$. Any coordinate $\boldsymbol{s} \in \mathcal{S}_K \setminus \mathcal{S}_{K \cap I}$ satisfies $|\mathrm{supp}(\boldsymbol{s})| \leq |K| \leq k$ and contains at least one index $i \in K \setminus I$ in its support, so $\boldsymbol{s} \in \mathcal{R}_i^k$ for such an i. Hence,

$$W_{\mathcal{S}_K \setminus \mathcal{S}_{K \cap I}}(g) \leq \sum_{i \in K \setminus I} \mathrm{Inf}_i^k[g] < |K \setminus I| \cdot \frac{\epsilon}{k} \leq \epsilon,$$

where the strict inequality follows from the definition of I (the bound holds trivially if $K \subseteq I$). □

Based on these lemmas, we are now in a position to present the main result.

Theorem A.2. There exists an algorithm such that, for any d-dimensional feature space $\mathbf{\Omega}$ with maximum alphabet size c, any binary classifier f over $\mathbf{\Omega}$, and any product distribution $\mathcal{D}$ with full support over $\mathbf{\Omega}$, given any size constraint $k \leq d$, parameters $\epsilon, \delta \in (0, 1)$, and membership query access to f, the algorithm returns a subset $K \subseteq [d]$ of size at most k satisfying, with probability at least $1 - \delta$:

$$\mathsf{err}_{f,\mathcal{D}}(K) \leq \min_{K' \subseteq [d], |K'| \leq k} \mathsf{err}_{f,\mathcal{D}}(K') + \epsilon.$$

The algorithm is fixed-parameter tractable in k, c, and $1/\epsilon$, with a runtime polynomial in d, $(ck/\epsilon)^k$, and $\ln(1/\delta)$.

Proof. Let K be the subset returned by Algorithm 2, with $\epsilon' = 2\epsilon/7$ and $\theta = \epsilon' / c^{k/2}$, and let K^* maximize $W(f_{K^*})$ among subsets of size at most k. By Lemma A.4, K^* minimizes the global relevance error among such subsets, and

$$\mathsf{err}_{f,\mathcal{D}}(K) - \mathsf{err}_{f,\mathcal{D}}(K^*) = \frac{1}{2} \left(W(f_{K^*}) - W(f_K) \right),$$

so it suffices to show that $W(f_{K^*}) - W(f_K) \leq 2\epsilon$. By Theorem A.1, with probability at least $1 - \delta$, the IKM phase returns a hypothesis g with $L_\infty(f - g) \leq \theta$ and satisfying the properties of Proposition A.2. We assume this event in what follows, and decompose the energy gap as:

$$\begin{aligned} W(f_{K^*}) - W(f_K) = &\underbrace{W(f_{K^*}) - W(g_{K^*})}_{\text{(A)}} + \underbrace{W(g_{K^*}) - W(g_{K^* \cap I})}_{\text{(B)}} \\ &+ \underbrace{W(g_{K^* \cap I}) - W(g_K)}_{\text{(C)}} + \underbrace{W(g_K) - W(f_K)}_{\text{(D)}}. \end{aligned}$$

By Lemma A.5 with ϵ' in place of ϵ, the recovery errors (A) and (D) are each bounded by $3\epsilon'$. By Lemma A.6 with the threshold ϵ'/k, the pruning error (B) is bounded by ϵ'. Since K maximizes $W(g_{K'})$ over subsets $K' \subseteq I$ of size at most k, and $K^* \cap I$ is such a subset, the optimality gap (C) is non-positive. Summing these terms yields

$$W(f_{K^*}) - W(f_K) \leq 3\epsilon' + \epsilon' + 0 + 3\epsilon' = 7\epsilon' = 2\epsilon.$$

For the runtime, we know by Theorem A.1 with $\theta = 2\epsilon/(7c^{k/2})$ that the IKM phase makes $O\big(dc^{3k+1}\epsilon^{-6} \ln \frac{dc^k}{\epsilon\delta}\big)$ queries and runs in time $O\big(dc^2 + d^2c^{3k+1}\epsilon^{-6} \ln \frac{dc^k}{\epsilon\delta}\big)$. It returns at most $L_0(g) \leq 4/\theta^2 = 49c^k/\epsilon^2$ coefficients, from which all restricted influences are computed in $O(d\, L_0(g))$ time.

To bound the size of I, we use the event of Proposition A.2, assumed above. On this event, every $\boldsymbol{s} \in \mathrm{supp}(g)$ satisfies $|\hat{g}(\boldsymbol{s}) - \hat{f}(\boldsymbol{s})| \leq \theta^2/4$, since all estimates are accurate, and $|\hat{f}(\boldsymbol{s})| \geq \theta/2$, by soundness. Hence $|\hat{g}(\boldsymbol{s})| \leq |\hat{f}(\boldsymbol{s})| + \frac{\theta}{2}|\hat{f}(\boldsymbol{s})| \leq \frac{3}{2}|\hat{f}(\boldsymbol{s})|$, using $\theta \leq 1$. By Parseval's identity, $W(g) \leq \frac{9}{4}\sum_{\boldsymbol{s}} \hat{f}(\boldsymbol{s})^2 = \frac{9}{4}$. Each coordinate $\boldsymbol{s}$ with $|\mathrm{supp}(\boldsymbol{s})| \leq k$ contributes to at most k restricted influences, so

$$\sum_{i=1}^{d} \mathrm{Inf}_i^k[g] \leq k\, W(g) \leq \frac{9k}{4}, \quad \text{and therefore} \quad |I| \leq \frac{9k/4}{\epsilon'/k} = \frac{63k^2}{8\epsilon}.$$

The exhaustive search examines all subsets of I of size at most k. Their number is at most $(e|I|/k)^k \leq (63ek/(8\epsilon))^k$ when $|I| \geq k$, and at most $2^{|I|} \leq 2^k$ otherwise; in both cases, it is at most $(63ek/(8\epsilon))^k$. Before the search, we keep only the coordinates of g whose support is included in I and has at most k elements, which takes $O(d\, L_0(g))$ time. Each energy $W(g_K)$ is then the sum of $\hat{g}(\boldsymbol{s})^2$ over the kept coordinates with support included in K, computed in $O(k\, L_0(g))$ time. The search thus takes $O\big(k\, L_0(g) \cdot (63ek/(8\epsilon))^k\big) = O\big(kc^k\epsilon^{-2}(63ek/(8\epsilon))^k\big)$ time. Adding the cost of the IKM phase and of the influence computation, the total runtime is polynomial in d, $(ck/\epsilon)^k$, and $\ln(1/\delta)$, which is fixed-parameter tractable in k, c, and $1/\epsilon$. □

B Experimental Appendix: Detailed Protocol and Extended Results

This appendix details the protocol of Section 5 and reports the complete results. The code, the raw results of every run, and the commands that produced them are provided in the supplementary material.

B.1 Parameters and Implementation of SE

Parameters. SE takes as input (ϵ, δ, k), the dimension d and the alphabet size c, and derives all its parameters as in Algorithm 2:

$$\epsilon' = \frac{2\epsilon}{7}, \qquad \theta = \frac{\epsilon'}{c^{k/2}}, \qquad \delta_0 = \frac{\theta^2\delta}{8dc}, \qquad b = \left\lceil 8 \ln \frac{1}{\delta_0} \right\rceil, \qquad n = \left\lceil \kappa \cdot \frac{64}{\theta^4} \right\rceil.$$

IKM retains a prefix when its estimated weight is at least $\theta^2/2$, and performs at most $N = 8dc/\theta^2$ estimations; the pruning step keeps the features i with $\mathrm{Inf}_i^k[g] \geq \epsilon'/k$. Every estimate is a median of b means of n samples. The constant κ is the only quantity that departs from the analysis, which requires $\kappa = 1$ (Lemma A.3). We calibrated it once, on hidden parities, as the smallest value with no failure over five seeds ($\kappa = 0.003$), and used $\kappa = 0.01$ throughout, which leaves a margin of about 3. For instance, with $\epsilon = 0.35$, $\delta = 0.1$, $c = 2$ and $k = 5$, we get $\theta \approx 0.0177$, $n = 6{,}553{,}600$ and b between 122 and 130 for $d \in [8, 20]$.

Implementation choices. Three choices depart from a literal reading of Algorithm 2 without affecting its guarantee.

- *Shared batch per depth.* At depth p, the weights of all prefixes of length p are estimated on the same b blocks of n samples, instead of one batch per prefix. As shown in Remark A.1, the guarantee is unchanged, and a run makes exactly $(2d+1)\, b\, n$ membership queries.

- *Degree cap.* IKM discards the prefixes whose support already holds more than k features. Both the restricted influences $\mathrm{Inf}_i^k[g]$ and the energies $W(g_K)$ with $|K| \le k$ only involve coordinates of degree at most k, so the discarded coordinates are never used.
- *Completion.* Since $K \subseteq K'$ implies $W(f_K) \le W(f_{K'})$, the error $\mathsf{err}_{f,\mathcal{D}}$ is non-increasing under inclusion. When fewer than k features survive the pruning, SE completes its explanation up to k features by decreasing estimated influence, at no additional query. Among all reported runs, this only happened on Bank Marketing (all k) and Breast Cancer ($k = 1$).

Machine cap. The implementation can cap the number of queries of a run at $r \cdot T$, where r is the query throughput measured on the target and T is a time limit (20 minutes by default); the cap lowers n, leaves b unchanged, and records the effective constant κ. Since the running time also grows with the number of prefixes estimated at each depth, this cap bounds the number of queries rather than the running time, which may exceed T. The cap was never reached in the runs reported in this paper, which all use $\kappa = 0.01$.

Software. The code, available at `https://github.com/Lounesbo/neurips-2026`, is written in `Python 3.14`. The classifiers are trained with `scikit-learn` (`MLPClassifier` with default hyperparameters), and their forward pass is reimplemented in `PyTorch`, so that membership queries are answered on the GPU in batches of up to 2^{22} instances.

Hardware. All experiments ran on a workstation with an Intel Xeon Gold 6252 CPU, 384 GB of RAM and two NVIDIA Quadro RTX 8000 GPUs (48 GB each), one run per GPU at a time.

B.2 Evaluation and Baseline

Exact evaluation. Every explanation K is scored by $\mathsf{err}_{f,\mathcal{D}}(K) = (1 - W(f_K))/2$. When $|\mathbf{\Omega}| \le 2^{22}$, which holds for all real-world benchmarks and all synthetic targets with $d \le 16$, the value $W(f_K)$ is computed exactly by enumerating $\mathbf{\Omega}$. For each assignment $\boldsymbol{a}$ of the features in K, the projection f_K takes the value $f_K(\boldsymbol{a}) = S_{\boldsymbol{a}}/\mathcal{D}_K(\boldsymbol{a})$, where $S_{\boldsymbol{a}} = \sum_{\boldsymbol{z}_K = \boldsymbol{a}} f(\boldsymbol{z})\,\mathcal{D}(\boldsymbol{z})$ and $\mathcal{D}_K(\boldsymbol{a}) = \prod_{i \in K} \mathcal{D}_i(a_i)$, so that $W(f_K) = \mathbb{E}[f_K^2] = \sum_{\boldsymbol{a}} S_{\boldsymbol{a}}^2/\mathcal{D}_K(\boldsymbol{a})$. The optimum opt is then computed by brute force over all subsets of size k (which suffices, by monotonicity), or in closed form for hidden parities. When $|\mathbf{\Omega}| > 2^{22}$ (the d-sweep with $d \ge 32$), $\mathsf{err}_{f,\mathcal{D}}(K)$ is estimated on 10^6 pairs of instances that agree on K.

Greedy baseline. GE implements forward selection [6]. At each step, it draws $5{,}000$ fresh instances, re-estimates the error of the current set K, and estimates the error of $K \cup \{i\}$ for every candidate i on an independent resample, so that the influence of one candidate cannot leak into the estimate of another. It stops when $|K| = k$ or when no candidate decreases the estimated error. Over all our experiments, a run of GE uses between $5 \cdot 10^4$ and $2 \cdot 10^6$ queries, against $4 \cdot 10^7$ to $4 \cdot 10^{10}$ for SE.

Runs and seeds. Each run of each explainer uses its own random generator, seeded by the series, the setting, and the run index, so that every run can be reproduced on its own. Synthetic settings are repeated 10 times (100 times for the δ-sweep), and real-world settings 6 times for each $k \in \{1, \dots, 5\}$.

B.3 Synthetic Targets: Confidence and Cost

Targets. The hidden parity is $\chi_S(\boldsymbol{x}) = \prod_{i \in S} (-1)^{x_i}$, with S drawn at random. For even c and uniform marginals, each factor has zero mean, so the whole energy of χ_S lies on coordinates with support S: $\mathsf{err}_{f,\mathcal{D}}(K) = 0$ if $S \subseteq K$ and $1/2$ otherwise. The noisy parity flips the label of χ_S ($d = 16$, $|S| = 3$) on a fixed pseudo-random subset of density $\eta = 0.05$, defined by hashing $\boldsymbol{x}$ so that f remains a function; its optimum, computed by brute force, is $\mathrm{opt} = 0.093 \approx 2\eta(1 - \eta)$. In the Synergy Trap ($d = 16$, $|S| = k - 1$), $\mathsf{err}_{f,\mathcal{D}}(\varnothing) = 3/8$ and $\mathsf{err}_{f,\mathcal{D}}(\{i^*\}) = 1/4$, while adding a proper subset of S leaves the error unchanged; only $\{i^*\} \cup S$ reaches $\mathsf{err}_{f,\mathcal{D}}(K) = 0$.

Confidence. Figure 2 reports the failure frequency of SE on the noisy parity, over 100 runs for each $\delta \in \{0.01, 0.02, 0.05, 0.1, 0.2, 0.5\}$. No run failed, and its gap to opt was zero in all 600 runs; the 95% Wilson intervals lie below δ for every $\delta \ge 0.05$. By contrast, GE succeeded in only 2% of these runs, with an average gap of 0.40.

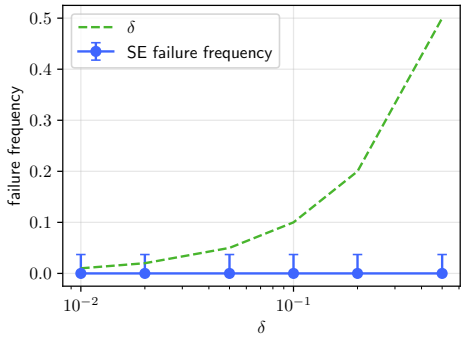


Figure 2: Failure frequency of SE against δ on the noisy parity ($d = 16$, $k = 3$, $\epsilon = 0.35$), with 95% Wilson intervals over 100 runs, and the bound δ (dashed).

Cost. Figure 3 reports the cost of SE on hidden parities, where one parameter varies and the others are fixed ($\epsilon = 0.35$, $\delta = 0.1$; $k = 3$ and $c = 2$ for the d-sweep; $d = 16$ and $c = 2$ for the k-sweep; $d = 16$ and $k = 2$ for the c-sweep). SE reached opt in all 130 runs, while GE succeeded in only 7 of the 120 runs with $|S| \geq 2$, as expected on parities. In every run, the number of queries is exactly $Q = (2d + 1)\, b\, n$ (left panels), where $n \propto \theta^{-4} = c^{2k}/\epsilon'^4$ and $b = \lceil 8 \ln(8dc/(\theta^2\delta)) \rceil$ grows only logarithmically. Table 1 (Section 5) decomposes the growth of Q along each sweep into these three factors. The number of weights and coefficients actually estimated by IKM (right panels) stays between 18 and 251 on average, several orders of magnitude below the bound $N = 8dc/\theta^2$, because the pruning at $\theta^2/2$ discards almost every prefix of a parity. The running time follows the same trends, from 1.6 s ($d = 8$) to 227 s ($d = 128$) at $k = 3$, and from 0.3 s ($k = 1$) to 93 s ($k = 5$) at $d = 16$.

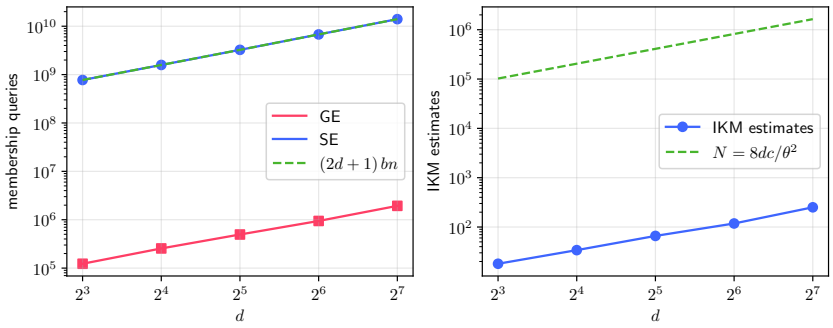
B.4 Real-World Benchmarks

Preprocessing. The nine datasets are binary classification tasks from `OpenML`. Instances with a missing label are removed, as are attributes with at least one missing value, and constant attributes. Each remaining attribute is binarized ($c = 2$): a numerical attribute with more than two values is split at its median ($x >$ median), an attribute with two values is kept as is, and a categorical attribute with more values is reduced to a test on its most frequent value. The data are split into 80% for training and 20% for testing, stratified by label. $\mathcal{D}$ is the product of the empirical marginals of the binarized training and test data, with Laplace smoothing, which ensures full support. Table 4 summarizes the datasets.

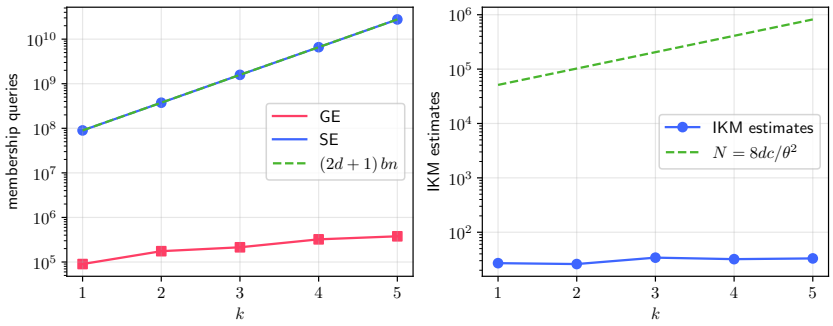
Table 4: Real-world benchmarks. Raw: attributes of the `OpenML` dataset; Cat./Num.: categorical and numerical attributes kept; Dropped: attributes removed because of missing values or a constant value. The binary dimension is $d =$ Cat. + Num.

				Attributes			Accuracy of f	
ID	Dataset	Instances	Raw	Cat.	Num.	Dropped	Train	Test
37	Pima Indians Diabetes	768	8	0	8	0	0.76	0.69
15	Breast Cancer (WI)	699	9	0	8	1	0.96	0.98
50	Tic-Tac-Toe	958	9	9	0	0	0.98	0.94
43,672	Heart Disease	1,190	11	0	11	0	0.86	0.79
179	Adult	48,842	14	9	2	3	0.83	0.83
42,192	COMPAS	5,278	13	6	7	0	0.68	0.62
1,461	Bank Marketing	45,211	16	9	7	0	0.89	0.89
24	Mushroom	8,124	22	20	0	2	1.00	1.00
31	Credit-G	1,000	20	13	7	0	0.86	0.70

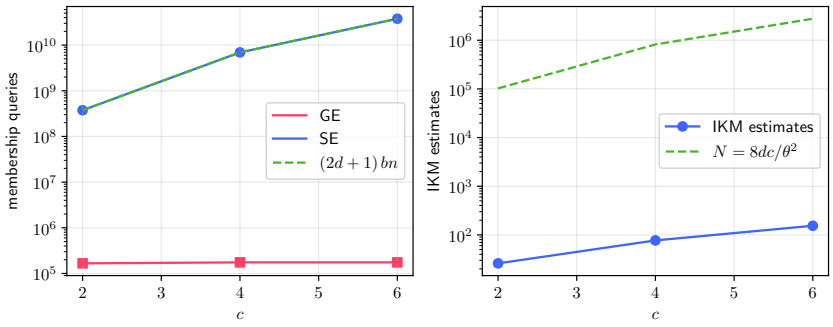
Complete results. Table 5 extends Table 2 to the nine datasets, and Figure 4 shows $\mathsf{err}_{f,\mathcal{D}}(K)$ against k. On Pima Indians Diabetes and Adult, both explainers reach opt in every run; on Heart Disease, GE misses it twice. On Bank Marketing, the label is strongly imbalanced, so that most of



(a) Dimension d ($k = 3$, $c = 2$)



(b) Explanation size k ($d = 16$, $c = 2$)



(c) Alphabet size c ($d = 16$, $k = 2$)

Figure 3: Cost of SE on hidden parities. Left: membership queries of SE and GE, with the value $(2d + 1)\,b\,n$ of the analysis (dashed). Right: number of estimations performed by IKM, with the bound $N = 8dc/\theta^2$ (dashed). Both axes of the d-sweep are logarithmic.

the energy of f lies on the constant coefficient: at most two features pass the pruning threshold, and SE relies on the completion for every k. It still reaches opt for $k \leq 3$, and misses it by 0.0016 and 0.0004 for $k = 4$ and $k = 5$, two near-ties like the other misses of SE. All runs of both explainers succeed. For all datasets, the curves of SE coincide with opt for most k, while those of GE show a spread across runs.

Explanations. Table 6 lists the explanations returned by SE at $k = 3$; each of them is optimal, and identical across runs. They are readable as such: for instance, on Adult, the prediction of the classifier is best explained by the education level, the marital status, and the capital gains of a person; on Tic-Tac-Toe, by the contents of the center and of two corners of the board.

Table 5: All real-world benchmarks ($\epsilon = 0.35$, $\delta = 0.1$); columns as in Table 2.

			opt reached		$\mathsf{err}_{f,\mathcal{D}}(K)$ at $k = 5$			
Dataset	d	Acc.	SE	GE	SE	GE	opt	Time (s)
Pima Indians Diabetes	8	0.69	30/30	30/30	0.075	0.075 ± 0.000	0.075	248
Breast Cancer (WI)	8	0.98	30/30	25/30	0.080	0.080 ± 0.000	0.080	244
Tic-Tac-Toe	9	0.94	30/30	21/30	0.249	0.254 ± 0.007	0.249	397
Heart Disease	11	0.79	30/30	28/30	0.176	0.176 ± 0.000	0.176	722
Adult	11	0.83	30/30	30/30	0.076	0.076 ± 0.000	0.076	580
COMPAS	13	0.62	30/30	26/30	0.110	0.112 ± 0.003	0.110	949
Bank Marketing	16	0.89	18/30	13/30	0.047	0.049 ± 0.002	0.047	843
Mushroom	20	1.00	24/30	17/30	0.183	0.185 ± 0.003	0.183	1998
Credit-G	20	0.70	24/30	21/30	0.217	0.221 ± 0.004	0.217	2464
Total			246/270	211/270				

Table 6: Explanations returned by SE at $k = 3$, all equal to the optimum. Each feature is denoted by the binary test that defines its binarized attribute. An explanation is a set of features, not a conjunction of literals: it partitions $\mathbf{\Omega}$ into the $2^3 = 8$ regions defined by the outcomes of its three tests.

Dataset	Explanation K returned by SE ($k = 3$)	$\mathsf{err}_{f,\mathcal{D}}(K)$
Pima Indians Diabetes	plasma glucose > 117, body mass index > 32, age > 29	0.158
Breast Cancer (WI)	clump thickness > 4, cell shape uniformity > 1, bland chromatin > 3	0.139
Tic-Tac-Toe	top-left = x, middle-middle = x, bottom-left = x	0.326
Heart Disease	sex $= 1$, chest pain type ≥ 4, exercise angina $= 1$	0.288
Adult	education-num > 10, marital status = married-civ-spouse, capital gain $= 0$	0.124
COMPAS	age > 31, priors count > 2, age category = less than 25	0.138
Bank Marketing	housing loan = yes, call duration > 180, previous outcome = unknown	0.051
Mushroom	odor = none, gill size = broad, stalk surface above ring = smooth	0.235
Credit-G	checking status = no checking, duration > 18, savings < 100	0.247

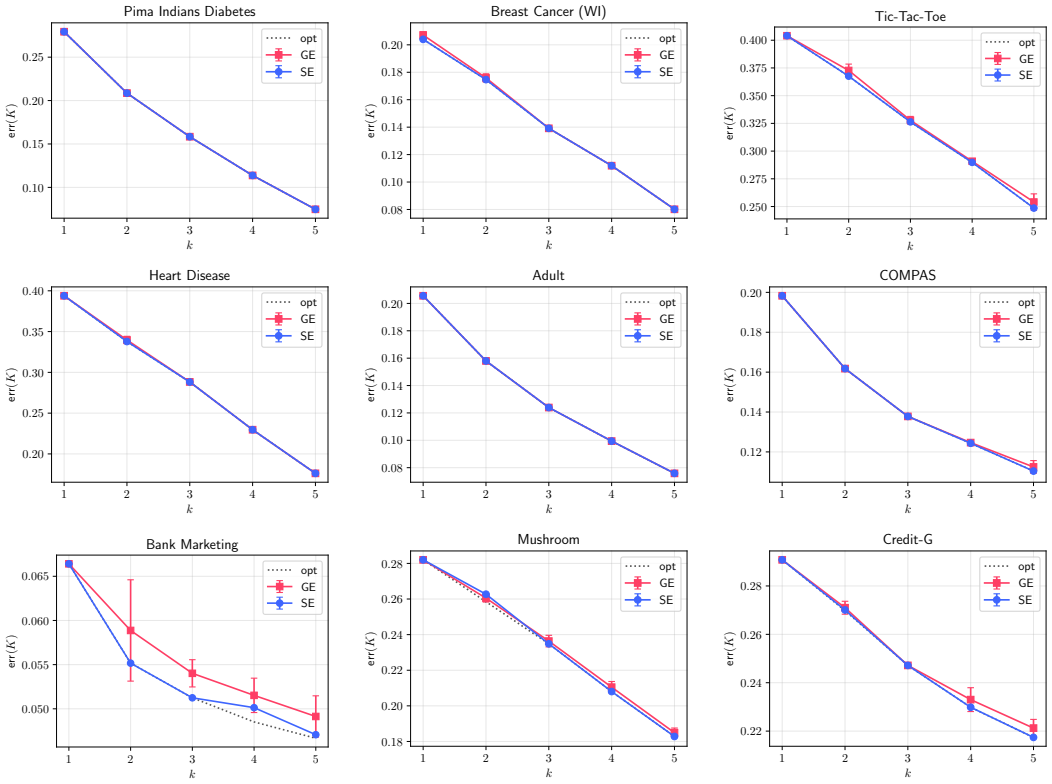


Figure 4: $\mathrm{err}_{f,\mathcal{D}}(K)$ against k on the real-world benchmarks, with opt (dotted). Bars show the range over 6 runs.

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